arXiv:2607.25468v1 [cs.CV] 28 Jul 2026

Safety-Aware Cascaded Inference for Crop Damage Assessment with Controlled Error Trade-offs

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Abstract

In picture-based agricultural insurance for smallholder farmers, missed damage detections carry substantially higher cost than false alarms: a farmer who sustained real losses receives no payout, while unnecessary expert review is operationally costly but reversible. Standard multi-class classifiers optimize global accuracy but provide no mechanism to operationalize or control this asymmetric cost structure at inference time.

We propose CascadeCropNet, a two-stage cascade architecture that can be calibrated to satisfy a target recall constraint (Rec-Damaged ≥ 0.95) under a chosen operating point through threshold selection. A lightweight Sentinel model performs binary health triage; samples exceeding a calibrated damage probability threshold τ are escalated to a specialist Expert model for fine-grained diagnosis. This design provides explicit, deployment-time control over the safety–efficiency trade-off without retraining.

Evaluated on the Eyes on the Ground dataset (28,077 images, 23,804 retained after label consolidation, from Kenyan smallholder maize farms), the cascade achieves Rec-Damaged = 0.974 at $\tau = 0.5$, reducing missed

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Submitted to Computers and Electronics in Agriculture, April 2026

damage cases by up to 54% at a selected operating point relative to a flat baseline. Under evaluation alignment, the representational gap reduces to +0.008 F1-macro, confirming that the contribution is architectural rather than representational. Under input degradation, the system prioritizes escalation over confident misclassification, reflecting error containment through architectural isolation rather than intrinsic model robustness.

These results demonstrate that cascade architectures can operationalize and control safety-oriented decision constraints through calibrated routing in deployment settings where reliability and controllability matter more than aggregate accuracy. These properties depend on threshold calibration and deployment conditions and do not constitute guarantees under arbitrary distribution shift.
Keywords: crop damage assessment, cascaded inference, safety-constrained learning, smallholder agriculture, picture-based insurance

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1. Introduction

Crop damage assessment at scale is fundamentally a decision problem, and like most decision problems that matter, it has an asymmetric cost structure. When an automated system fails to detect damage on a smallholder maize plot, the farmer who sustained real losses receives no payout. When it flags a healthy crop as damaged, an agronomist reviews an unnecessary case. These two errors are not equivalent. The first has consequences that are difficult to reverse; the second is operationally recoverable. Any system deployed in an insurance-linked agricultural workflow that does not encode this asymmetry into its design risks solving a misaligned problem. In East African picture-based insurance schemes covering smallholder maize farms, this misalignment has direct financial consequences: a system that misclassifies a damaged plot as healthy triggers no payout for a farmer who may have no other safety net [1, 2].

This is not a new observation. Cost-sensitive learning has formalized the asymmetric error problem since at least [3]. Neyman-Pearson classification frames it as constrained risk minimization — maximize performance on one class subject to a hard bound on the error rate of another [4, 5]. Selective classification offers a related response: abstain when confidence is insufficient, and defer to a human [6]. The theoretical apparatus exists. What has received less attention is how these ideas translate into deployed systems operating under real distribution shift — where the asymmetric objective encoded at training time does not reliably produce asymmetric behavior when inputs degrade [7, 8]. Importantly, the robustness demonstrated in this work is not a system-wide property. Rather, it arises from architectural constraints that selectively stabilize the diagnostic component, conditional on correct routing decisions at the triage stage. That gap is what this paper addresses. The specific setting is crop damage classification from smartphone field images, using the Eyes on the Ground dataset [9]: 23,804 georeferenced images from smallholder maize farms across eight Kenyan counties, collected under a picture-based insurance protocol [10] and validated by agronomists. The images are exactly as difficult as that description implies — variable lighting, heterogeneous devices, inconsistent framing, the full range of field conditions that make controlled-environment benchmarks a poor guide to deployment reality [11, 12]. On this data, a flat multi-class classifier trained with standard objectives — the V11 baseline described in Section 4.6 — achieves F1-Damage of 0.777 and F1-Growth of 0.748 under clean conditions (Section 6.1). Under input corruption — sensor noise, color shift, blur — the same model's diagnostic performance degrades by 17 to 26 percent across tasks simultaneously (Section 6.5), consistent with the absence of architectural separation between corruption-sensitive and corruption-invariant components [7].

The system proposed here, CascadeCropNet, is a two-stage cascade in which the first stage — SentinelNet — performs binary health triage under a training-time safety constraint, and the second stage — ExpertNet — performs fine-grained damage classification exclusively on samples routed as damaged. The safety constraint is explicit: Rec-Damaged ≥ 0.95, enforced at the level of checkpoint selection across training epochs (Section 4.2). Checkpoint selection maximizes precision on damaged samples subject to this bound — a constrained optimization surrogate for the operational problem of minimizing unnecessary expert consultations while maintaining a hard floor on missed damage cases [3, 4]. This is not merely a post-hoc threshold adjustment; the constraint is enforced during model selection itself.

The routing threshold τ is a deployment-time parameter. It navigates a Pareto frontier between expert load ρ — the fraction of samples routed to ExpertNet — and Rec-Damaged. Operators can move along this frontier without retraining: at $\tau = 0.5$, the system achieves Rec-Damaged $= 0.974$ with a 13.2% reduction in expert inference volume; at $\tau = 0.6$, Rec-Damaged $= 0.929$ with a 22.6% reduction (see Section 6.4). The selected checkpoints satisfy the safety constraint independently of the operating point τ.

The more important property is structural. ExpertNet always receives images drawn from a clean preprocessing pipeline, regardless of the corruption applied to Sentinel inputs. Under field and sensor corruption conditions, ExpertNet's diagnostic metrics — F1-Expert, F1-DGT, F1-WED — remain stable within measurement precision, because the applied corruption does not propagate to its inputs (Section 6.5). What changes is ρ: the Sentinel, uncertain under degraded inputs, routes more samples to the Expert. The system fails toward consultation rather than toward confident errors. The flat baseline has no such mechanism; corruption propagates through all tasks simultaneously and performance degrades across the board [7].

This is the paper's central claim, stated precisely: we identify a mismatch between objective-level risk encoding and system-level behavior under distribution shift, and show that enforcing decision structure at the architectural level provides a practical mechanism to mitigate this mismatch, yielding error-containment and decision-structure properties not reliably achieved through loss function design alone under distribution shift [13, 8]. More broadly, this suggests that under distribution shift, decision structure may need to be enforced at the system level rather than inferred from objective functions alone.

The contributions are as follows.

Architectural. A safety-aware cascaded inference framework that separates triage from diagnosis under asymmetric error costs, restructuring system failure modes by localizing missed detections at the triage stage rather than eliminating them globally.

Operational. A threshold-based routing mechanism enabling deployment-time control of the safety–efficiency trade-off through calibrated τ selection without retraining. The trade-off structure between Rec-Healthy and Rec-Damaged is monotonic and predictable across evaluated settings, making recalibration tractable.

Analytical. A decomposition of system behavior (Experiments 1–5) showing that architectural isolation yields error containment rather than intrinsic model robustness: ExpertNet's diagnostic stability under input corruption arises from pipeline design, not learned invariance, and the representational gap between cascade and flat baseline reduces to +0.008 F1-macro under fair evaluation.

Empirical. Evidence on the Eyes on the Ground agricultural dataset demonstrating reduced missed damage cases (up to 54% at $\tau = 0.5$ relative to a flat baseline) and stable expert performance under Sentinel-input corruption, with explicitly characterised trade-offs between safety, efficiency, and input quality.

The paper is organized as follows. Section 2 reviews related work on crop health assessment, multi-task learning, constrained classification, and cascade architectures. Section 3 describes the Eyes on the Ground dataset and the label consolidation decisions made for this work. Section 4 presents the CascadeCropNet architecture and training protocol. Section 5 defines the evaluation protocol. Section 6 reports results, including the threshold sweep and robustness stress test. Sections 7 and 8 discuss implications and limitations. Section 9 concludes.

2. Related Work

Automated crop health assessment from images has a clear origin story, and it is worth being honest about what that story actually shows. [14] demonstrated that a deep convolutional network trained on the PlantVillage dataset [15] — 54,306 images of diseased and healthy leaves, photographed against controlled backgrounds — could classify 26 diseases across 14 crop species with accuracy exceeding 99%. The result was widely cited. It was also, in retrospect, a ceiling that real-world deployment had no realistic chance of reaching. The PlantVillage images are clean, centered, uniformly lit, and collected under conditions that bear essentially no resemblance to what a smallholder farmer's smartphone produces in the field. [11] confronted this directly in their work on cassava disease detection in Tanzania: accuracy dropped substantially when the same modeling approach was applied to field-collected imagery, and the gap between controlled-environment performance and deployment performance became one of the defining methodological problems of the subfield. Subsequent work — including [16] on the limits of deep learning for plant disease detection, [17] on transfer learning strategies for plant pathology, and [12] on mobile capture device-based crop disease classification in field conditions — has probed this gap with varying success, but has not closed it under realistic deployment conditions [18, 19]. This suggests that the limitation is not only one of model capacity, but of mismatch between training conditions and deployment environments [20].

The Eyes on the Ground dataset [9] is positioned squarely in the hard regime. Images were captured by smallholder farmers across eight Kenyan counties using standard smartphone cameras, following a picture-based insurance protocol described by [10]. Lighting, zoom, angle, and device quality vary substantially across the collection. Labels were assigned by agronomists or trained professionals, with automatic labels generated by an in-house algorithm where manual annotation was unavailable. This combination of real-world image noise and expert-validated labels is what makes the dataset appropriate for safety-critical training — you need both properties, and most agricultural imaging datasets reliably provide at most one [21, 22, 23].

Multi-task learning is the natural framework when a single image carries multiple semantically related labels. The foundational argument, due to [24], is that shared representations across related tasks improve generalization by providing auxiliary supervision signal. More recent work has extended this to uncertainty-weighted loss formulations [25] and attention-based task coordination [26], with [27] providing a comprehensive overview of the landscape. The question of which tasks benefit from joint learning — and which do not — has been studied empirically by [28], who show that task groupings matter as much as architecture. In agricultural imaging, multi-task approaches have been used to jointly predict disease presence and severity [29], growth stage and damage type [29], and various combinations of phenological and disturbance labels. The standard architecture — shared backbone, task-specific heads, summed loss — treats all tasks symmetrically. This is the right design when tasks are equally important and their labels are independent. It is the wrong design when one task label is only semantically meaningful conditional on another. Damage type — Drought vs. Weeds — is only a relevant prediction given that the crop is damaged. Training a damage-type head on healthy samples introduces noise: the head receives gradient signal from cases where its output is undefined by the label structure. While sufficiently expressive models may learn to approximate this conditional structure implicitly, doing so requires capacity to be spent modeling label dependencies that can instead be enforced directly [28]. Alternative approaches, such as learned gating or conditional computation within a single model, can approximate this structure, but do not guarantee that irrelevant samples are excluded from expert processing at inference time. Masked supervision, where the expert head is conditioned on a damage indicator and receives gradient only from damaged samples, is the principled fix. [30] and [31] have shown in related hierarchical label settings that conditional supervision consistently outperforms symmetric multi-task training when label dependencies are strong. The cascade proposed here extends this logic from training to inference by enforcing conditional structure at the level of input distribution rather than gradient flow: ExpertNet never processes samples the Sentinel classified as Healthy, eliminating the problem architecturally rather than managing it through loss weighting.

The literature on cost-sensitive learning and constrained classification is older and more rigorous than its uptake in applied agricultural ML would suggest. [3] showed that cost-sensitive classification can be reduced to cost-insensitive classification with reweighted examples [32] — a result that is clean theoretically but does not address what happens when the input distribution shifts at deployment. The problem of learning from imbalanced data, closely related to cost-sensitive learning, has been surveyed by [33] and addressed through techniques including synthetic oversampling [34], threshold calibration [35], and re-weighting [36]. Neyman-Pearson classification [4, 5] frames the problem as constrained risk minimization: maximize recall on one class subject to a hard bound on the error rate of the other. This closely matches the objective structure used in the Sentinel's training, implemented through constraint-aware model selection rather than direct optimization: maximize precision on damaged samples subject to Rec-Damaged ≥ 0.95. Selective classification [6] offers a related but structurally different response: when confidence is insufficient, abstain and defer to a human. The distinction matters. Abstention removes decisions from the system entirely, reducing coverage [37]. Cascade routing preserves nominal coverage while redistributing error modes rather than eliminating them — uncertain cases are not rejected but escalated to a specialist within the system boundary. These are different system philosophies, and conflating them obscures what the cascade actually does.

In picture-based insurance schemes, the cost asymmetry is structural rather than parametric. A false negative results in claim denial for a farmer who sustained real losses — a harm that cannot be reversed after the assessment window closes. A false positive results in unnecessary expert review — an operational cost that scales with volume but does not cause irreversible harm to any individual farmer. The ratio of these costs depends on the specific insurance product and regional context [2, 1, 10], but their ordering is invariant: in any insurance scheme where payouts are contingent on assessment, denial of a legitimate claim is categorically worse than triggering unnecessary review.

Cascaded classifiers have a long history in computer vision, beginning with the Viola-Jones face detector [38], where a sequence of increasingly complex classifiers filtered easy negatives early, reducing total computation dramatically. Convolutional cascade approaches [39] extended this to deep feature representations. The deep learning era has revisited the cascade idea primarily through the lens of efficiency: adaptive computation networks [40], early-exit architectures [41], multi-exit networks [42], dynamic routing [43], spatially adaptive computation [44], and multi-scale dense networks [45] all exploit the intuition that easy examples should not consume the same compute budget as hard ones [46]. What these systems optimize is throughput. What the Sentinel optimizes is safety. The routing criterion is not confidence in any prediction — it is the probability of damage exceeding a threshold calibrated to a recall constraint. This reframes the cascade from a computational efficiency device into a safety enforcement mechanism. This distinction matters because optimizing for efficiency implicitly assumes symmetric error costs, whereas safety-constrained routing explicitly encodes asymmetric risk [3, 4]. To our knowledge, safety-constrained cascades of this form — where routing is governed by a training-time recall guarantee rather than a confidence heuristic — have received limited attention in the agricultural imaging literature.

Robustness under distribution shift is a well-documented failure mode for convolutional networks generally [7, 47, 8] and agricultural imaging systems specifically [11, 12]. [48] showed that ImageNet-trained networks are biased toward texture features that are particularly vulnerable to certain corruption types. Corruption types that occur naturally in field deployment — Gaussian noise from low-quality sensors, motion blur from handheld capture, color shift from device heterogeneity — reliably degrade performance, often more than reported benchmark gaps would suggest [7]. The standard response is data augmentation [49]: expose the model to corrupted inputs during training and hope the learned representation generalizes. This works to a degree. What it cannot reliably do is protect one component of a multi-task system from corruption affecting another, because in a flat architecture all tasks share the same representation and therefore the same vulnerability. The reliability of uncertainty estimates under such shift is itself questionable [13, 50]. Architectural isolation — structurally separating the component that processes potentially corrupted inputs from the component that performs fine-grained diagnosis — has received comparatively little explicit treatment as a robustness mechanism, particularly in settings where robustness to input corruption is a primary objective rather than a secondary effect. The robustness results reported in Section 6 follow directly from this separation, not from any augmentation strategy applied to ExpertNet.

3. Dataset

The Eyes on the Ground dataset [9] was produced through a collaboration between ACRE Africa, the International Food Policy Research Institute, the Lacuna Fund, KALRO, BlueGreen Labs, and Dvara E-Registry, and is hosted on Radiant MLHub under a CC-BY-SA-4.0 license (DOI: 10.34911/rdnt.1bs2jw). It contains nearly 28,077 georeferenced and timestamped crop images collected from smallholder maize farms across eight counties in Kenya during the 2020–2021 growing seasons. After label consolidation and exclusion of classes with insufficient samples or excessive visual overlap (nutrient deficiency, wind, disease, flooding, pest, and animal damage), 23,804 images are retained for training, validation, and testing. Images were captured by farmers using standard smartphone cameras following the picture-based insurance protocol of [10], in which farmers photograph their plots at regular intervals throughout the growing season as a condition of insurance participation [2, 1]. Geolocation was recorded via mobile phone GPS; exact coordinates are not released, and spatial references are reported only at the village level using GADM36 bounding boxes to protect contributor privacy [9].

The label taxonomy is rich. Each image carries annotations along three dimensions: crop damage category, phenological growth stage, and damage extent. Damage categories include DGT (drought), WED (weed), WND (wind), DSE (disease), FLD (flooding), PS (pest), ANM (animal), ND (nutrient deficit), and GC (good crop). Growth stages span sowing, vegetative, flowering, and maturity. The full taxonomy is documented in the dataset property table of [9], which lists both manual ML labels (`growth_stage`, `damage`, `extent`) and extended probabilistic labels (`drought_probability`, `disturbance_weed`, `disturbance_drought`, and others) derived from independent automatic classifiers. All labels were assigned by agronomists or trained professionals where manual annotation was feasible; an in-house machine learning algorithm supplied automatic labels where manual review was unavailable, with automatic label probabilities derived from independent models and therefore not constrained to sum to unity. This labeling pipeline is imperfect by design — it reflects the operational realities of large-scale data collection in the field — and the mixed manual-automatic provenance should be understood as a property of the dataset, not a defect to be corrected away. Because automatic labels are produced by independent models, the resulting noise is structured rather than purely random [30], which may affect both class boundaries and calibration [50]. The impact of this mixed provenance is implicitly evaluated through the robustness and generalization behavior reported in Section 6.

For this work, the full label taxonomy is consolidated into three binary or categorical targets aligned with the cascade's two-stage decision structure. Health status collapses to a binary label: Healthy (GC) versus Damaged (DGT + WED). This restriction defines a controlled setting for evaluating asymmetric decision policies under class imbalance [33, 35]; evaluating the cascade under the full multi-class taxonomy is an important direction for future work. Nutrient deficiency cases are excluded from the damage severity task — the class contains too few samples for reliable supervised learning and preliminary inspection indicates substantial visual overlap with drought stress, making inclusion more likely to introduce label ambiguity than useful signal [51]. Growth stage is consolidated from four phenological classes to two: Early (Sowing + Vegetative) and Late (Flowering + Maturity). This aggregation may discard finer-grained temporal cues, but provides a more stable supervisory signal given class frequencies, while preserving the agronomically meaningful distinction between vegetative and reproductive phases.

The cascade uses a hierarchical label schema with three targets — `bin_label`, `expert_label`, `growth_label` — while the flat baseline uses a 3-class damage label (Healthy, Drought, Weeds) alongside `growth_label`. These two schemas are applied to the same underlying image collection but encode different assumptions about task structure, and comparison between systems trained under each schema requires care.

The dataset is partitioned into train, validation, and test splits at a 70/15/15 ratio using fixed CSV files, with splits performed at the image level and verified to have no overlap across sets (Table 1). Because plot-level identifiers are not available, independence at the field level cannot be confirmed; this may introduce residual correlation across splits due to repeated observations of the same plots over time [10], and reported performance metrics should be interpreted with this limitation in mind. Splits are stratified on the damage label to preserve class frequencies. The resulting distribution is summarized in Table 1. The training set contains 16,662 images; validation and test sets contain 3,571 each. Three levels of class imbalance are present and require explicit treatment [33, 34], all of which are visible in the class distribution plots of Figure 1. At the health level, 73.3% of images are Damaged and 26.7% are Healthy — a roughly 2.7:1 ratio that persists consistently across all three splits (Figure 1, left panel). Among damaged images, Weeds account for 79.9% of cases and Drought for 20.1%, a 4:1 imbalance that represents the primary challenge for the Expert model (Figure 1, center panel). At the growth stage level, Late growth (Flowering + Maturity) accounts for 69.9% of images versus 30.1% Early — a milder imbalance that requires no special weighting beyond standard cross-entropy (Figure 1, right panel).

These three imbalances are not independent problems to be managed separately. They interact. The Sentinel faces a dual challenge: identifying the minority Healthy class while maintaining a hard recall constraint on the majority Damaged class [4, 5]. The Expert must distinguish Drought from Weeds in a 4:1 imbalanced regime, on a subset of images that has already been filtered by the Sentinel's routing decision. Imbalance handling

Table 1: Dataset split statistics. Health, damage type, and growth stage distributions are reported as percentages within each split. Source: [9]; split proportions computed from the stratified 70/15/15 partition described in Section 3.

Split	**Total**	**Healthy**	**Damaged**	**Drought**	**Weeds**	**Early**	**Late**
Train	16,662	26.7%	73.3%	14.8%	58.5%	30.1%	69.9%
Val	3,571	26.8%	73.2%	14.7%	58.5%	28.8%	71.2%
Test	3,571	26.7%	73.3%	14.8%	58.5%	29.7%	70.3%
Total	**23,804**	26.7%	73.3%	14.8%	58.5%	29.9%	70.1%

Table 2: SentinelNet (V13) and flat baseline (V11) training configuration.

Setting	**V13 Sentinel**	**V11 Baseline**
Epochs	30	20
Batch size	32	16
Learning rate	1e-4	1e-4
Weight decay	0.01	0.01
LR schedule	ReduceLROnPlateau	ReduceLROnPlateau
Patience/factor	3 / 0.5	3 / 0.5
Backbone warmup	3 epochs	5 epochs
Class weights	[1.37, 3.70]	None
Selection criterion	max(Safety Score)	max(0.6·F1-D + 0.4·F1-G)

Table 3: ExpertNet (V14) training configuration.

Setting	**V14 Expert**
Epochs	25
Batch size	16
Learning rate	1e-4
Weight decay	0.05
LR schedule	CosineAnnealingLR
Backbone warmup	5 epochs
Progressive sampler	Yes
Corruption augmentation	ControlledCorruption + RandomNoise (50% clean, 20% blur, 30% colour jitter, $\sigma = 0.08$, $p = 10\%$)
Selection criterion	max(0.7·F1-Exp + 0.3·Rec-H)

is therefore specific to each stage: the Sentinel uses inverse-frequency class weights of [1.37, 3.70] for Damaged and Healthy respectively, combined with ClassBalancedFocalLoss [52]; the Expert uses FocalLoss with $\gamma = 2.0$ [53] on the damage-type head, combined with a progressive sampler that upweights damaged samples during training. These choices follow standard practice for imbalanced classification [33, 36] and are applied consistently across models to reduce confounding from class imbalance, although differences in optimization dynamics across architectures may still persist [25]. Growth stage prediction at both stages uses standard binary cross-entropy without reweighting, given the milder imbalance. The full training configuration for all three models is described in Section 4.

One property of this dataset deserves emphasis beyond the class statistics. The images are genuinely difficult. Smartphone cameras vary in sensor quality, lens characteristics, and automatic exposure behavior — a consequence of the mobile phone GPS collection protocol documented in [9]. Farmers photograph their plots from varying distances and angles. Lighting conditions range from direct midday sun — which washes out color information — to overcast conditions that flatten contrast. Motion blur from handheld capture is common [49]. These are not pathological edge cases; they are the baseline distribution [7], and therefore constitute a natural domain shift relative to controlled imaging conditions [47, 8]. The stress test reported in Section 6 evaluates robustness under this shift directly, using the three corruption conditions visualized in Figure 7.

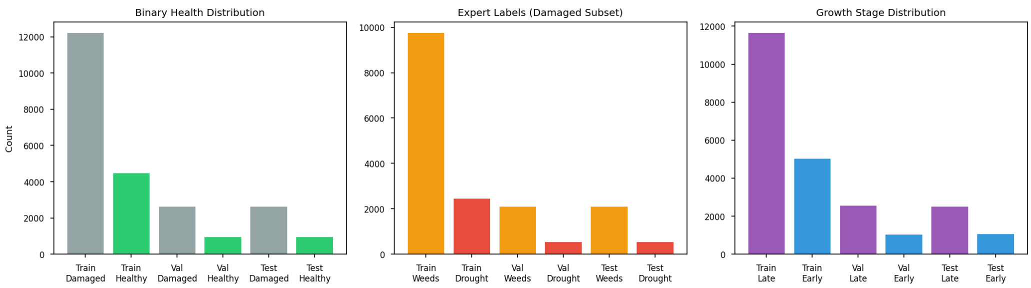

Figure 1: Class distribution across train, validation, and test splits. **Left:** Binary health distribution (Damaged vs. Healthy). The 73.3% Damaged rate is consistent across all three splits, reflecting stratified sampling on the damage label. **Center:** Expert label distribution within the damaged subset. Weeds (WED) account for 79.9% of damaged samples versus 20.1% Drought (DGT), producing a 4:1 class imbalance addressed via FocalLoss [53] in ExpertNet training. **Right:** Growth stage distribution (Late vs. Early). Late growth (Flowering + Maturity) accounts for 69.9% of all images. Data from [9].

4. Method

The design of CascadeCropNet is motivated by a single observation: the decision structure of crop damage assessment is hierarchical by nature, and a flat classifier that ignores this structure pays a cost in both safety and robustness [28, 7]. Damage type is only a meaningful prediction conditional on damage presence. Growth stage is an auxiliary signal relevant throughout. These are not implementation details — they are properties of the label space that should be reflected in the architecture, not managed through loss weighting after the fact [24, 27].

4.1. Design Rationale

We cast crop damage triage as a constrained decision problem, implemented via constrained model selection over training checkpoints. Let $\mathbf{x}$ denote an input image and $y \in \{\text{Healthy, Damaged}\}$ its binary health label. The Sentinel learns a scoring function $f(\mathbf{x}) = P(\text{Damaged}|\mathbf{x})$, and a threshold τ is selected on the validation set such that the resulting classifier maximizes Safety Score subject to Rec-Damaged ≥ 0.95. Formally:

$$\tau^* = \arg\max_{\tau} \text{ Safety Score}(\tau) \quad \text{s.t.} \quad \text{Rec-Damaged}(\tau) \geq 0.95 \qquad (1)$$

Precision serves as a tractable proxy for operational efficiency — minimizing unnecessary expert consultations — while the recall constraint encodes the asymmetric cost floor: the cost of missing damage exceeds the cost of unnecessary review [3, 4]. This formulation can be interpreted as enforcing a minimum safety constraint while optimizing operational efficiency within the feasible region — a form of structured risk control in which the safety floor is non-negotiable and efficiency is maximized subject to it [5, 6]. This is not a full cost model [32]. It is a principled surrogate that makes the asymmetry explicit and enforceable at the level of model selection rather than gradient optimization.

Expert load ρ — the fraction of samples routed to ExpertNet — is reported directly as the primary efficiency metric, alongside Rec-Damaged as the primary safety metric. The choice of precision as the secondary objective reflects its operational alignment: a Sentinel that produces fewer false positives routes fewer healthy samples to the Expert, reducing ρ directly. Precision and ρ are related but not identical; reporting ρ explicitly avoids conflating the two [54]. The constraint and objective together define a Pareto frontier, with x-axis expert load ρ and y-axis Rec-Damaged defining the operating space. The threshold τ navigates this frontier at deployment time without retraining, as visualized in Figure 5 (left panel).

4.2. SentinelNet

SentinelNet (V13) is the first stage of the cascade. Its primary task is binary health discrimination; its routing output determines which samples ExpertNet processes. The backbone is MobileNetV3-Large [55] pretrained on ImageNet [56], with approximately 4.2 million parameters. Input resolution is 224×224 pixels. The shared feature extractor feeds into a fully connected layer ($960 \rightarrow 1024$) followed by batch normalization and ReLU activation, from which three task-specific heads branch: a binary health head ($1024 \rightarrow 256 \rightarrow 2$, Healthy/Damaged), an auxiliary damage-type head ($1024 \rightarrow 256 \rightarrow 2$, Drought/Weeds), and an auxiliary growth stage head ($1024 \rightarrow 256 \rightarrow 2$, Early/Late). All heads use ReLU activation and Dropout(0.2) before the final linear layer.

The training loss combines three terms with explicit weighting:

$$\mathcal{L}_{\mathrm{S}} = 5 \cdot \mathcal{L}_{\mathrm{bin}}^{\mathrm{CBFocal}} + 1 \cdot \mathcal{L}_{\mathrm{exp}}^{\mathrm{CE}} + 0.1 \cdot \mathcal{L}_{\mathrm{gro}}^{\mathrm{CE}} \tag{2}$$

The binary health loss uses ClassBalancedFocalLoss [52] with inverse-frequency class weights [1.37, 3.70] for Damaged and Healthy respectively, derived from the training set frequencies reported in Table 1. The heavy weighting on the binary term ($\times 5$) reflects the primacy of health triage in the cascade's decision structure.

The auxiliary growth stage head contribution to the total loss is zeroed after the first five training epochs ($\mathcal{L}_{\text{gro}} = 0$ for epoch ≥ 5), concentrating gradient signal on the binary health and auxiliary damage-type tasks as training progresses [27]. The auxiliary damage-type head receives standard cross-entropy supervision throughout all 30 training epochs at weighting coefficient 1.0, providing continuous signal for the shared representation to encode damage-relevant visual features [24].

Training runs for 30 epochs with AdamW [57] (lr $= 1 \times 10^{-4}$, weight decay $= 0.01$), gradient clipping at max_norm = 1.0, and ReduceLROnPlateau scheduling (patience = 3, factor = 0.5). The backbone is frozen for the first three epochs with batch normalization in evaluation mode to stabilize early feature extraction [55]. We define *Safety Score* as the precision on the Damaged class evaluated at the highest-confidence threshold satisfying both Rec-Damaged ≥ 0.95 and Rec-Healthy ≥ 0.40 simultaneously. Epochs where no such threshold exists receive Safety Score = 0 and are ineligible for selection. Checkpoint selection maximizes Safety Score on the validation set at each epoch. The threshold τ is tuned per epoch on the validation set by sweeping $P(\text{Damaged}|\mathbf{x})$ over a dense grid and selecting the value that maximizes the Safety Score subject to the constraints in Equation 1. This mechanism enforces satisfaction of the recall constraint on the validation set, avoiding approximation errors introduced by surrogate penalty terms [4, 5]. Training dynamics are shown in Figure 3, showing loss curves, recall trajectories for Rec-Healthy and Rec-Damaged, and the optimal τ evolution across 30 epochs.

4.3. ExpertNet

ExpertNet (V14) is the second stage. It receives only samples routed by the Sentinel and performs fine-grained classification between Drought and Weeds. The architecture mirrors SentinelNet — MobileNetV3-Large backbone [55], shared fully connected layer, three task-specific heads — but operates at higher input resolution (256 × 256 pixels) to capture the finer visual detail that distinguishes drought stress from weed competition [58, 18]. The damage-type head is the primary output; the binary health head and growth head serve as auxiliary tasks. Training dynamics for V14 are shown in Figure 4, showing loss curves, per-class diagnostic F1 scores (F1-DGT and F1-WED), and the composite HP-Score across 25 epochs.

The training loss weights the damage-type task heavily:

$$\mathcal{L}_{\mathrm{E}} = 1 \cdot \mathcal{L}_{\mathrm{bin}}^{\mathrm{CE,weighted}} + 3 \cdot \mathcal{L}_{\mathrm{exp}}^{\mathrm{Focal}(\gamma=2)} + 0.1 \cdot \mathcal{L}_{\mathrm{gro}}^{\mathrm{CE}} \tag{3}$$

FocalLoss with $\gamma = 2.0$ [53] on the damage-type head addresses the 4:1 Weeds/Drought imbalance documented in Section 3 by down-weighting easy majority-class examples during training [52]. A progressive sampler further upweights damaged samples in each batch [33]. Corruption augmentation is applied during training — ControlledCorruption (50% clean, 20% Gaussian blur, 30% colour jitter) combined with RandomNoise ($\sigma = 0.08$, applied with probability 10%) [49] — to improve the Expert's robustness to inputs that may arrive with residual preprocessing artifacts. This augmentation is applied to ExpertNet's training distribution, not to Sentinel inputs at inference.

Training runs for 25 epochs with AdamW [57] (lr $= 1 \times 10^{-4}$, weight decay $= 0.05$) and CosineAnnealingLR scheduling. The backbone is frozen for the first five epochs. Checkpoint selection maximizes the composite score $0.7 \cdot$ F1-Expert $+ 0.3 \cdot$ Rec-Healthy on the validation set, with the best checkpoint identified at epoch 7 (combined_hp = 0.8278). Total training time across all three models is approximately 20.7 hours: 4.2 hours for V11, 10.0 hours for V13 SentinelNet, and 6.5 hours for V14 ExpertNet, all on GPU with a fixed random seed of 42 for reproducibility.

4.4. Cascade Routing and Architectural Isolation

At inference, the cascade operates as follows. Every input image is first processed by SentinelNet at 224×224 pixels, producing $P(\text{Damaged}|\mathbf{x})$. If $P(\text{Damaged}|\mathbf{x}) \geq \tau$, the image is routed to ExpertNet at 256×256 pixels; the Expert's damage-type output (Drought or Weeds) is used as the final prediction. If $P(\text{Damaged}|\mathbf{x}) < \tau$, the image is predicted Healthy and ExpertNet is never invoked. Growth stage prediction is always drawn from the Sentinel, regardless of routing. The fraction of samples routed to ExpertNet is ρ, the expert load. This inference procedure is illustrated in Figure 5, which also shows the full threshold sweep across $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ on the validation set.

We define *architectural isolation* formally as follows: ExpertNet's input distribution is invariant to corruptions applied to Sentinel inputs, under a fixed preprocessing pipeline. This holds structurally. Corruption affects $P(\text{Damaged}|\mathbf{x})$ and therefore ρ — the routing composition — but does not affect the pixel values passed to ExpertNet, which are always drawn from the clean preprocessing pipeline applied to the original source image. Architectural isolation therefore guarantees *input quality invariance*, not *selection stability*. These are two distinct properties [13]. Under corruption, the set of samples reaching ExpertNet changes — the Sentinel routes more aggressively when uncertain — but each routed sample arrives at ExpertNet with its representation intact. This routing-induced shift may also alter the conditional difficulty of the expert task, as samples routed under high

Sentinel uncertainty are not drawn from the same distribution as clean training examples [8]. The robustness results reported in Section 6 follow from the input quality invariance property; the deployment caveat about routing-induced distribution shift follows from the selection stability property.

This distinction from abstention-based approaches is structural, not incidental [37, 6]. Selective classifiers reduce coverage by removing low-confidence decisions from the system. The cascade preserves nominal coverage while redistributing error modes rather than eliminating them [4]: every input receives a final prediction, but the decision pathway depends on the Sentinel's confidence. Uncertain samples are escalated within the system rather than deferred outside it.

4.5. Masked Supervision

The damage-type head in both SentinelNet and ExpertNet is supervised exclusively on damaged samples. Formally, the expert loss term is conditioned on the damage indicator:

$$\mathcal{L}_{\text{exp}} = \mathbf{1}_{[y \in \{\text{Drought, Weeds}\}]} \cdot \ell\big(f_{\text{exp}}(\mathbf{x}),\, y_{\text{exp}}\big) \tag{4}$$

where ℓ is the appropriate loss function and f_{exp} is the damage-type head output. Without this masking, the expert head receives gradient signal from healthy samples — cases where the damage-type label is structurally undefined — introducing inconsistent supervision signals for samples where the damage-type label carries no meaning [28, 30]. Masked supervision enforces the conditional structure of the label space at the gradient level; the cascade enforces it at the input distribution level at inference. These two mechanisms are complementary and operate at different stages of the learning and inference pipeline [27, 26].

4.6. Flat Baseline

The flat baseline (V11, BaselineNetV11) uses the same MobileNetV3-Large backbone [55] with two task-specific heads: a 3-class damage head (Healthy, Drought, Weeds) and a binary growth head (Early, Late). Input resolution is 224×224 pixels. The training loss is an equal-weighted sum of cross-entropy losses on both tasks. No routing, no safety constraint, no masked supervision. The selection criterion maximizes $0.6 \cdot$ F1-Damage $+ 0.4 \cdot$ F1-Growth on the validation set. Training runs for 20 epochs with AdamW [57] (lr $= 1 \times 10^{-4}$, weight decay $= 0.01$) and ReduceLROnPlateau scheduling, with a five-epoch backbone warmup, converging at epoch 9 as shown in Figure 2. The baseline is deliberately minimal: its purpose is to isolate the effect of the cascade architecture and safety constraint against a comparable backbone [55]. The absence of stronger flat baselines should be interpreted as a limitation of the current study rather than evidence of superiority over all flat approaches.

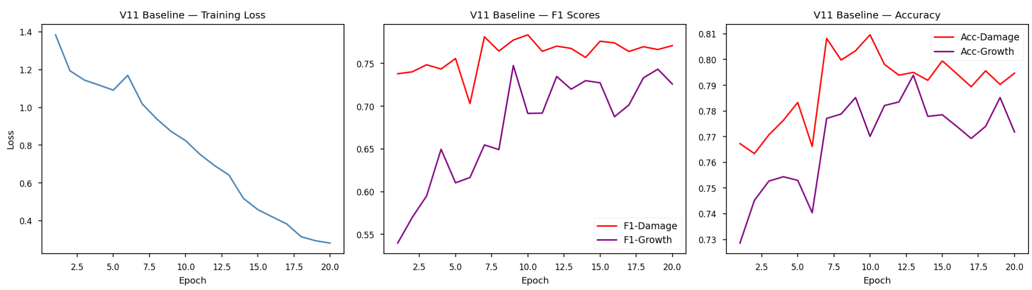

Figure 2: V11 flat baseline training dynamics across 20 epochs. **Left:** Training loss decreasing smoothly, converging by epoch 9. **Center:** F1-Damage (3-class macro) and F1-Growth (binary macro) trajectories. F1-Damage plateaus around 0.77–0.78 from epoch 9 onward, consistent with capacity limits of a flat architecture attempting to model conditional label dependencies without structural enforcement [28]. **Right:** Accuracy on damage and growth tasks. Best checkpoint: epoch 9, combined score = 0.7654.

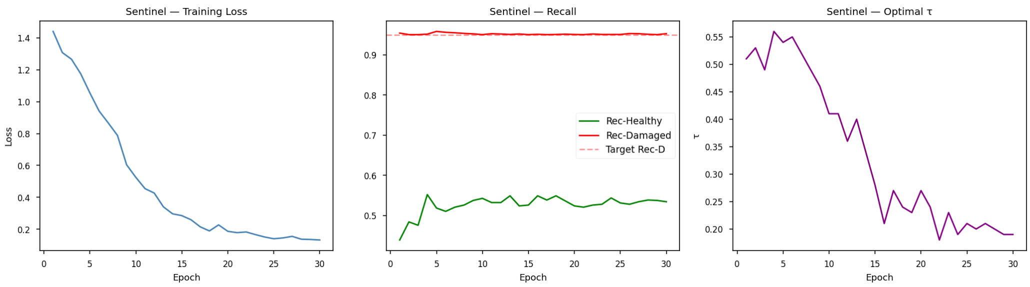

Figure 3: SentinelNet (V13) training dynamics across 30 epochs. **Left:** Training loss curve showing smooth convergence. **Center:** Recall trajectories for Rec-Damaged (red, maintained ≥ 0.95 throughout) and Rec-Healthy (green), with the target Rec-Damaged $= 0.95$ constraint shown as a dashed line. The safety constraint is satisfied at every epoch under appropriate threshold selection. **Right:** Optimal τ evolution across training epochs, drifting from $\tau = 0.56$ at epoch 4 to $\tau = 0.19$ at epoch 30, reflecting the model's increasing confidence in damage predictions. Best checkpoint: epoch 4, Safety Score = 0.8532.

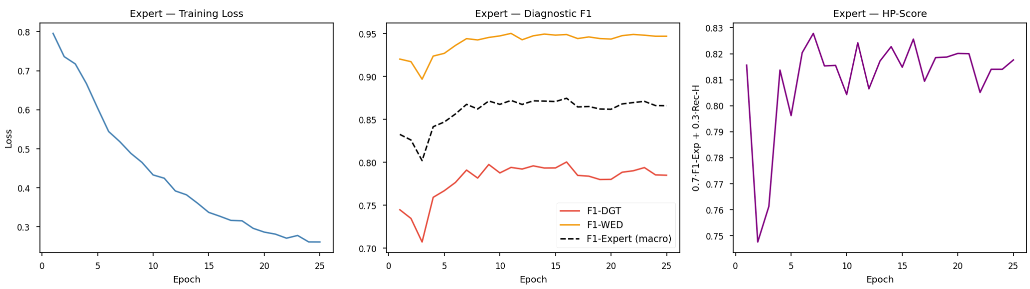

Figure 4: ExpertNet (V14) training dynamics across 25 epochs. **Left:** Training loss converging smoothly throughout. **Center:** Per-class diagnostic F1 scores. F1-WED (Weeds, orange) rises steeply and stabilizes near 0.95 by epoch 5. F1-DGT (Drought, red) climbs more slowly and plateaus near 0.79, consistent with the 4:1 Weeds/Drought imbalance [53, 52] and feature-space overlap between the two damage classes [51]. F1-Expert macro (dashed black) stabilizes around 0.87. **Right:** Composite HP-Score ($0.7 \cdot$ F1-Expert $+$ $0.3 \cdot$ Rec-Healthy) across epochs. Best checkpoint: epoch 7, combined_hp = 0.8278.

Reproducibility

All experiments use a fixed random seed of 42 applied to Python `random`, NumPy, PyTorch, and `cudnn.deterministic=True`. The framework is PyTorch with torchvision, using MobileNetV3-Large pretrained on ImageNet [56, 55]. Dataset splits are fixed CSV files with no overlap between train, validation, and test sets. Thresholds $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ are evaluated post-hoc on the validation set after training is complete; threshold selection never touches the test set. Training flags default to checkpoint loading (`TRAIN_FLAT`, `TRAIN_SENTINEL`, `TRAIN_EXPERT` all default `False`). Total training time across all three models is approximately 20.7 hours: 4.2h for V11, 10.0h for V13 SentinelNet, and 6.5h for V14 ExpertNet, all on GPU. Model checkpoints are saved with `model_state_dict`, threshold, Safety Score, and epoch metadata. The complete training configuration is documented in Tables 2 and 3. Code and checkpoints are available from the corresponding author upon reasonable request.

5. Evaluation Protocol

Evaluation in a two-stage cascade requires choices that a flat classifier does not face. The most consequential is this: should ExpertNet be evaluated on samples that the Sentinel actually routes to it, or on the full set of truly damaged samples? These are not the same population. In deployment, only sentinel-routed samples reach ExpertNet — the Expert never sees images the Sentinel classified as Healthy, regardless of their true label. Evaluating on routed samples would confound two distinct sources of error: Sentinel routing mistakes and ExpertNet diagnostic mistakes. Separating them requires a deliberate protocol choice [54, 59].

Throughout this work, ExpertNet is evaluated on truly damaged samples

— the full set of images carrying a Drought or Weeds label in the ground truth, irrespective of how the Sentinel routed them. This isolates ExpertNet's intrinsic diagnostic capacity from triage performance, enabling controlled comparison of what the Expert actually learned. For completeness, evaluation under actual Sentinel routing is also captured in the end-to-end cascade metrics — Rec-Damaged, Rec-Healthy, F1-bin, and ρ — ensuring that both intrinsic and deployed Expert performance are observable. End-to-end cascade performance constitutes the primary evaluation, while decomposed Expert metrics serve diagnostic interpretation. The trade-off is transparency about what each level measures: F1-Expert, F1-DGT, and F1-WED reflect the Expert's representational quality; end-to-end metrics reflect system behavior under actual routing. In deployment, if routing is systematically biased — over-routing certain crop varieties, under-routing early-stage damage — deployed Expert performance will differ from the decomposed figures reported here [13]. This separation assumes that representation quality generalizes across routing-induced distribution shifts [8]. That assumption remains empirically unverified and constitutes an open problem for follow-on work.

The threshold sweep is conducted on the validation set ($n = 3{,}571$) across six operating points: $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$, as described in Section 4.1. At each τ, expert load ρ and all cascade metrics are computed. The Pareto frontier between ρ (x-axis) and Rec-Damaged (y-axis) is visualized in Figure 5 (left panel), alongside the full metric sweep across τ values in Figure 5 (right panel). While both the cascade and the flat baseline are subject to validation-dependent model selection, the cascade introduces an additional operating parameter τ, which may increase sensitivity to selection effects relative to V11 [35]. This asymmetry is acknowledged; reported metrics should be interpreted accordingly.

The robustness stress test is conducted on the held-out test set ($n = 3{,}571$) under three input corruption conditions applied exclusively to Sentinel inputs. ExpertNet always receives images from the clean preprocessing pipeline, consistent with the architectural isolation property defined in Section 4.4. The three conditions are: **Clean** (standard resize and normalize), **Field** (ColorJitter with brightness and contrast factor 0.4, combined with GaussianBlur with $\sigma \in [0.1, 2.0]$), and **Sensor** (additive Gaussian noise with $\sigma = 0.08$ applied post-normalization), as described in the Reproducibility subsection above. These conditions simulate the dominant corruption types present in the Eyes on the Ground dataset [9]: device heterogeneity and exposure variation (Field), and sensor noise from low-quality smartphone cameras (Sensor) [7, 49]. Corruption is applied to Sentinel inputs only — not to ExpertNet inputs — because this reflects the intended deployment design: in a real pipeline, the Expert model operates on images that have passed through a controlled server-side preprocessing stage, while the Sentinel operates closer to the raw capture device [10]. The flat baseline, which has no such architectural separation, is evaluated under identical corruption conditions applied uniformly to its single model. This asymmetry is not an evaluation artifact — it is a direct consequence of the architectural difference being tested.

The cascade decomposes a single harder joint classification problem into two simpler conditional ones [28]. A reviewer might reasonably ask whether performance gains simply reflect an easier subtask rather than a genuine system improvement. The response is structural: while each subtask is individually simpler, the overall system is evaluated on the full input distribution — every image receives a final prediction, and end-to-end metrics capture the complete decision chain including routing errors. Task decomposition is the architectural contribution; the end-to-end evaluation is what makes the gains meaningful [3, 4].

A note on split usage. The threshold sweep uses the validation set; the stress test uses the test set. τ is a validation-time hyperparameter — selecting it on the test set would constitute evaluation leakage [35] — while the stress test is a post-hoc robustness evaluation applied to a fixed, already-selected model. The practical consequence is that threshold sweep metrics and stress test metrics are not directly comparable as absolute numbers across splits. Both sets of results are reported with their respective split origins made explicit. Harmonizing these evaluations to a single held-out split is a methodological improvement left for future work, as noted in Section 8.

The following metrics are reported throughout this work. **Rec-Damaged**: recall on the Damaged class, the primary safety metric and the quantity constrained during Sentinel training [4, 5]. **Rec-Healthy**: recall on the Healthy class, the explicit operational cost of the safety floor [3]. **F1-bin**: macro F1 on the binary health task at the cascade level, capturing end-to-end routing performance [59]. **F1-Expert**: macro F1 on the damage-type task (Drought vs. Weeds), computed on truly damaged samples. **F1-DGT** and **F1-WED**: per-class F1 scores for Drought and Weeds respectively, drawn from the Expert's confusion matrix as shown in Figure 6. **F1-Growth**: macro F1 on the binary growth stage task, always predicted by the Sentinel [59]. ρ: expert load, the fraction of samples routed to ExpertNet, reported as the primary efficiency metric. All F1 scores are macro-averaged [54, 59]. Confusion matrices for all three tasks at $\tau = 0.5$ are provided in Figure 6. Training dynamics for all three models are documented in Figures 2, 3, and 4 for V11, V13 SentinelNet, and V14 ExpertNet respectively, with Figure 1 providing class distributions across splits for reference.

6. Results

6.1. Flat Baseline

The flat baseline (V11) converges at epoch 9 of 20, with the combined selection criterion ($0.6 \cdot \text{F1-Damage} + 0.4 \cdot \text{F1-Growth}$) reaching 0.7654. F1-Damage — reported as 3-class macro across Healthy, Drought, and Weeds — is 0.777. Accuracy on the damage task is 0.803. F1-Growth is 0.748, with accuracy 0.785. Training curves are shown in Figure 2: loss decreases smoothly through epoch 9 and the F1-Damage score plateaus around 0.77–0.78 for the remainder of training, indicating that the model extracts most available signal from the joint 3-class objective early and does not meaningfully improve thereafter. This plateau is consistent with the capacity limits of a flat architecture attempting to simultaneously model health status, damage type, and growth stage from a single shared representation without structural enforcement of the conditional label dependencies described in Section 3 [28, 24].

6.2. Safety Performance

The Sentinel (V13) reaches its best checkpoint at epoch 4, with a Safety Score (defined in Section 4.2) of 0.8532. The threshold at this checkpoint is $\tau = 0.56$. Training dynamics are shown in Figure 3: Rec-Damaged remains above 0.95 at every epoch throughout the full 30-epoch training run, confirming that the safety constraint is satisfied at the level of checkpoint selection consistently and not only at the selected checkpoint [4]. The threshold drifts from 0.56 at epoch 4 to 0.19 at epoch 30, reflecting the model's increasing confidence in damage predictions as training progresses [50]. The Rec-Damaged ≥ 0.95 floor reflects the operational requirement that at most 5 missed damage cases per 100 truly damaged plots are tolerable, consistent with the irreversibility of claim denial in picture-based insurance [2, 10].

At $\tau = 0.5$, the cascade achieves Rec-Damaged = 0.974 on the validation set. The test set contains approximately 2,616 truly damaged samples. At the flat baseline's Rec-Damaged of 0.943, approximately 148 damaged cases are missed. At the cascade's Rec-Damaged of 0.974, approximately 68 cases are missed — a reduction of 80 missed damage cases, corresponding to a 3.1 percentage point improvement and roughly 54% fewer missed damage cases in absolute terms. Experiment 1 (Section 6.7) confirms this reduction is not explained by threshold calibration of the flat baseline alone. In an insurance-linked workflow [2, 1], each missed damage case represents a farmer who sustained real losses and received no payout. That reduction is not a metric abstraction.

Rec-Healthy at $\tau = 0.5$ is 0.424. This is low, and it should be stated plainly rather than explained away. The Sentinel is trained under a constraint that prioritizes damage recall over healthy precision — the safety floor is Rec-Damaged ≥ 0.95, not Rec-Healthy $\geq$ any bound beyond the minimum 0.40 imposed during checkpoint selection [3, 4]. The model does exactly what it was trained to do: it aggressively routes toward damage detection, accepting a high false-alarm rate on healthy crops as the explicit operational cost of the safety constraint. At $\tau = 0.6$, Rec-Healthy improves to 0.650 with Rec-Damaged = 0.929, below the 0.95 target but within a range that may be acceptable in lower-risk deployment contexts [60]. The τ parameter gives operators direct control over this trade-off without retraining [6, 41].

The Rec-Damaged ≥ 0.95 floor is one point on a continuous trade-off curve. Lower constraints reduce the safety floor but increase the pool of feasible checkpoints and the achievable Safety Score. Higher constraints reduce feasible checkpoints — at Rec-Damaged ≥ 0.99, very few epochs may satisfy the joint constraint, potentially yielding Safety Score = 0 for all epochs. The 0.95 value was selected as a conservative operational floor consistent with the irreversibility of false negatives in insurance assessment; operators with different risk tolerances may adjust this constraint without retraining by modifying the checkpoint selection criterion and rerunning checkpoint evaluation on locally representative validation data.

6.3. Expert Diagnostic Quality

ExpertNet (V14; Section 4.3) reaches its best checkpoint at epoch 7, with the composite selection score ($0.7 \cdot$ F1-Expert + $0.3 \cdot$ Rec-Healthy) of 0.8278. F1-Expert — macro F1 across Drought and Weeds, computed on truly damaged samples — is 0.868. Per-class scores are F1-WED = 0.944 and F1-DGT = 0.791. Training dynamics are shown in Figure 4: F1-WED converges rapidly and stabilizes above 0.94 by epoch 5; F1-DGT climbs more slowly and shows greater variance, consistent with the minority class status of Drought (20.1% of damaged samples) and the 4:1 class imbalance [53, 52, 33]. The composite HP-Score stabilizes around 0.82 from epoch 10 onward.

The Drought result requires direct engagement. F1-DGT = 0.791 means that on the test set's approximately 527 drought-damaged samples, roughly 110 cases are misclassified — either predicted as Weeds or, at the Sentinel stage, missed entirely before reaching the Expert. FocalLoss with $\gamma = 2.0$ [53] partially addresses the 4:1 Weeds/Drought imbalance by down-weighting easy majority examples, but does not resolve it. The gap between F1-WED = 0.944 and F1-DGT = 0.791 is not a rounding error — it reflects a real representational challenge [51, 18]. Drought stress and weed competition produce visually overlapping symptoms in maize at certain growth stages: leaf yellowing, stunted growth, and canopy thinning occur in both conditions [21, 22]. At F1-DGT = 0.791, the system may not be sufficient for deployment settings that require treatment-level reliability, where distinguishing drought-appropriate responses from weed management interventions has direct agronomic consequences [10, 2].

A note on comparability. V11's F1-Damage of 0.777 is a 3-class macro score computed across Healthy, Drought, and Weeds on the full evaluation set. The cascade's F1-Expert of 0.868 is a 2-class macro score computed on truly damaged samples only — healthy samples never reach the Expert and are excluded from this metric by design. These metrics measure different things and a direct numerical comparison between them is not methodologically valid [54, 59]. The cascade's higher expert-class score reflects in part the task decomposition: the Expert solves a simpler conditional problem rather than the harder joint problem the flat model faces [28]. This decomposition is the architectural contribution, not a confound to be corrected. It follows directly from the conditional label structure established in Section 3: damage type is only meaningful given damage presence, and the cascade enforces this structure at inference [24, 27]. A fair evaluation of whether the Expert's representational quality exceeds the flat model's on the same samples would require restricting the baseline evaluation to damaged samples — an analysis not conducted here and noted as a methodological gap. Specifically, renormalizing V11's output probabilities over the damage classes $\{P(\mathrm{DGT}|\mathbf{x}), P(\mathrm{WED}|\mathbf{x})\}$ on the truly damaged subset would test whether the learned representation retains discriminative capacity when the healthy class is removed at inference time — a test of representation reuse under evaluation alignment, not a test of equivalent training conditions: V11 was trained on all samples including healthy crops, while ExpertNet was trained exclusively on damaged samples with progressive upweighting.

6.4. Threshold Sweep

The cascade is evaluated across six operating points $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ on the validation set ($n = 3{,}571$). Results are reported in Table 4 and visualized in Figure 5. The left panel of Figure 5 shows the Pareto frontier between expert load ρ (x-axis) and Rec-Healthy (y-axis), with each τ value marked explicitly. The right panel shows Rec-Damaged, Rec-Healthy, and F1-bin as functions of τ. Expert load ρ serves as a proxy for the activation rate of the damage decision region, quantifying how input corruption or threshold variation shifts the effective routing boundary under a fixed model.

Two properties of this sweep deserve explanation before the numbers are read. F1-Expert and F1-Growth are invariant to τ across all six operating points — both remain at 0.868 and 0.828 respectively regardless of threshold setting. This is a property of the evaluation protocol, not of the architecture. ExpertNet is evaluated on the full truly-damaged set at every operating point, independent of what the Sentinel actually routes; growth prediction always comes from the Sentinel. In deployment, where only routed samples reach ExpertNet, both metrics would depend on routing composition and would vary with τ [13, 8]. Readers should interpret the τ-invariant Expert and Growth scores as measures of intrinsic model quality, not as predictions of deployed system behavior.

At $\tau = 0.3$, $\rho = 0.973$ — nearly the entire dataset is routed to the Expert, and Rec-Damaged $= 0.997$ at the cost of Rec-Healthy $= 0.095$. The safety guarantee is maximized but the efficiency benefit is eliminated. At $\tau = 0.8$, $\rho = 0.500$ — half the dataset bypasses the Expert — but Rec-Damaged drops to 0.660, a miss rate that is too high for insurance-linked deployment [2, 1]. The operating point $\tau = 0.5$ provides Rec-Damaged $= 0.974$ with a 13.2% reduction in expert inference volume under clean input conditions, satisfying

Table 4: Cascade threshold sweep on the validation set ($n = 3{,}571$) across $\tau \in \{0.3, \ldots, 0.8\}$. Expert load ρ denotes the fraction of samples routed to ExpertNet. F1-Expert and F1-Growth are τ-invariant under this evaluation protocol (see Section 5); both remain at 0.868 and 0.828 respectively across all operating points because ExpertNet is evaluated on the full truly-damaged set independent of routing, and growth prediction always comes from the Sentinel.

τ	ρ	**Expert Reduction**	**Rec-H**	**Rec-D**	**F1-bin**	**F1-Exp**	**F1-Gro**
0.3	0.973	2.7%	0.095	0.997	0.515	0.868	0.828
0.4	0.932	6.8%	0.233	0.992	0.623	0.868	0.828
0.5	0.868	13.2%	0.424	0.974	0.730	0.868	0.828
0.6	0.774	22.6%	0.650	0.929	0.804	0.868	0.828
0.7	0.657	34.3%	0.816	0.830	0.795	0.868	0.828
0.8	0.500	50.0%	0.936	0.660	0.719	0.868	0.828

the safety constraint. This efficiency gain is conditional: it collapses to zero under severe input degradation as $\rho \to 1$ (Section 7.3) [41, 45]. The operating point $\tau = 0.6$ provides the best F1-bin of 0.804 with a 22.6% expert load reduction, at the cost of Rec-Damaged = 0.929 — below the 0.95 training target but potentially acceptable under constrained expert bandwidth where some safety margin can be traded for operational efficiency [6]. Section 7.1 discusses the deployment implications of this trade-off.

6.5. Robustness Under Distribution Shift

This result demonstrates *error containment via architectural decoupling* rather than general model robustness: because corruption at the Sentinel stage cannot propagate to ExpertNet's inputs, ExpertNet's input distribution does not change and its diagnostic outputs do not change.

The stress test is conducted on the test set ($n = 3{,}571$) under three

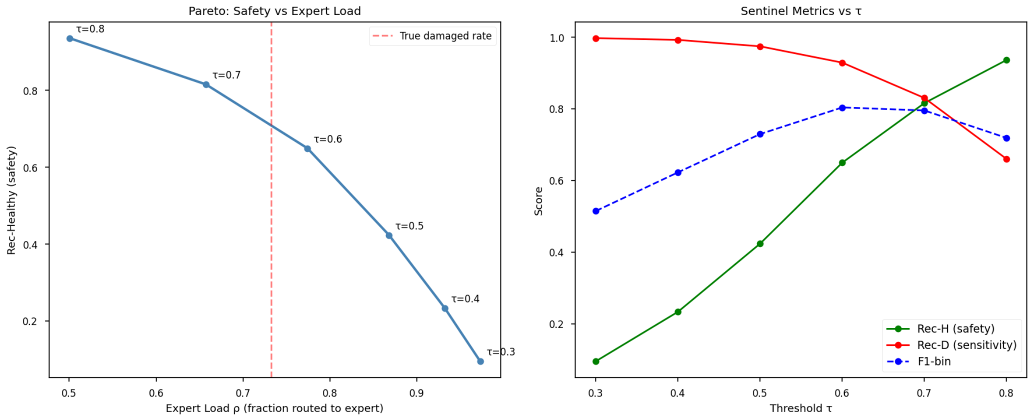

Figure 5: Cascade threshold sweep and Pareto frontier on the validation set ($n = 3{,}571$). **Left:** Pareto frontier between expert load ρ (x-axis, fraction routed to ExpertNet) and Rec-Healthy (y-axis), with each τ value marked. The dashed vertical line indicates the true damaged rate (73.3%). Operating points $\tau = 0.5$ and $\tau = 0.6$ lie in the region of meaningful safety-efficiency trade-off. **Right:** Rec-Damaged (red), Rec-Healthy (green), and F1-bin (blue dashed) as functions of τ. Rec-Damaged falls steeply above $\tau = 0.6$, reaching 0.660 at $\tau = 0.8$ — below the threshold acceptable for insurance-linked deployment [2, 10].

conditions applied exclusively to Sentinel inputs: Clean, Field (ColorJitter + GaussianBlur), and Sensor (additive Gaussian noise $\sigma = 0.08$). ExpertNet always receives clean images. This design reflects the deployment assumption that Sentinel inputs arrive from heterogeneous field devices while ExpertNet operates under controlled server-side preprocessing [10, 9]. It creates an intentionally asymmetric comparison — cascade with partial corruption vs V11 with full corruption — and should be read as testing whether architectural separation delivers practical value under the stated assumption, not as a neutral worst-case benchmark. Experiment 7 (Section 6.7) examines the both-corrupted scenario. Results are shown in Figure 7 and Table 5.

Under all three conditions, ExpertNet's diagnostic metrics remain stable within measurement precision: F1-Expert = 0.877, F1-DGT = 0.805, F1-WED = 0.949 across Clean, Field, and Sensor conditions without variation. This stability is not a finding about model robustness in any general sense [47, 48] — it is a direct consequence of the isolation property. Corruption applied to Sentinel inputs does not propagate to ExpertNet's inputs under the fixed preprocessing pipeline; therefore ExpertNet's input distribution does not change and its outputs do not change. The observed robustness predominantly arises from this architectural property rather than from representational invariance learned during training [49]; whether limited intrinsic robustness also contributes under mild corruption is examined in Experiment 3 (Section 6.7).

What does change under corruption is ρ. Under Clean conditions, $\rho = 0.856$. Under Field corruption, ρ increases to 0.912. Under Sensor noise, ρ reaches 0.968 — nearly the entire dataset is routed to the Expert, eliminating the efficiency benefit entirely. The Sentinel, uncertain under degraded inputs [13], routes more aggressively. This is a conservative failure mode — the system errs toward expert consultation rather than toward confident misclassification [6] — but it has a practical cost: the efficiency gain of 13.2% at Clean conditions ($\rho = 0.856$) disappears under Sensor noise ($\rho = 0.968$). Under constrained operational settings where expert inference capacity is limited [10, 2], Sensor- level input degradation effectively reverts the system to full Expert routing, and the cascade provides no throughput advantage over running ExpertNet on all inputs. Operators deploying in environments with frequent sensor degradation should treat ρ as a function of input quality, not as a fixed operating parameter. Section 7.2 discusses the architectural scope and limits of this guarantee.

Table 5: Robustness stress test results on the test set ($n = 3{,}571$) under three corruption conditions applied exclusively to Sentinel inputs. ExpertNet always receives clean images. Expert-head metrics (F1-Expert, F1-DGT, F1-WED) remain stable within measurement precision across all conditions, consistent with the architectural isolation property (Section 4.4). V11 has no architectural isolation; all metrics degrade simultaneously [7].

Model	**Condition**	**Rec-H**	**Rec-D**	ρ	**F1-Exp**	**F1-DGT**	**F1-WED**	**F1-Gro**
Cascade	Clean	0.455	0.969	0.856	0.877	0.805	0.949	0.828
	Field	0.295	0.987	0.912	0.877	0.805	0.949	0.811
	Sensor	0.110	0.997	0.968	0.877	0.805	0.949	0.775
V11 Baseline	Clean	0.722	0.912	1.000	—	0.787	0.863	0.729
	Field	0.468	0.944	1.000	—	0.654	0.827	0.644
	Sensor	0.316	0.973	1.000	—	0.708	0.824	0.576

The flat baseline shows no such protection. Under Field corruption, V11's F1-DGT drops from 0.787 to 0.654 — a 17% relative degradation. F1-Growth drops from 0.729 to 0.644, a 12% relative degradation. Under Sensor noise, Rec-Healthy falls from 0.722 to 0.316 — a 56% relative drop — and F1-Growth degrades to 0.576, 21% below clean performance. These degradations occur simultaneously across all tasks because V11 has no architectural separation between corruption-sensitive and corruption-invariant components [7, 48].

Corruption enters the shared representation and propagates everywhere. The comparison is shown directly in Figure 7.

The cascade's advantage over V11 under stress grows with corruption severity. Under Field conditions, the cascade maintains F1-DGT = 0.805 against V11's 0.654 — a 23% relative advantage. Under Sensor noise, the cascade maintains F1-DGT = 0.805 against V11's 0.708 — a 14% relative advantage. For F1-Growth under Sensor noise, the cascade achieves 0.775 against V11's 0.576 — a 35% relative advantage. These gaps do not exist under Clean conditions because architectural isolation only confers advantage when corruption is present [8, 47]. Under Clean conditions the cascade's Expert metrics slightly exceed V11's damage-type metrics, but the comparison remains indirect given the 3-class versus 2-class distinction established in Section 6.3.

One important caveat. The stress test applies corruption to Sentinel inputs only, reflecting a deployment assumption where ExpertNet operates under controlled server-side preprocessing while Sentinel inputs arrive from heterogeneous field devices [10, 9]. If in practice the preprocessing pipeline itself degrades — due to transmission artifacts, lossy compression, or device-side preprocessing failures — the isolation guarantee weakens [13]. The robustness demonstrated here is real under the stated assumptions; it should not be extrapolated to arbitrary pipeline configurations.

6.6. Task Decomposition and the Comparison Question

The flat baseline's F1-Damage of 0.777 is a 3-class macro score computed over Healthy, Drought, and Weeds simultaneously [59]. The cascade's F1-Expert of 0.868 is a 2-class macro score computed on truly damaged samples only — healthy samples never reach the Expert and are excluded from this metric by design. These metrics measure different things and a direct

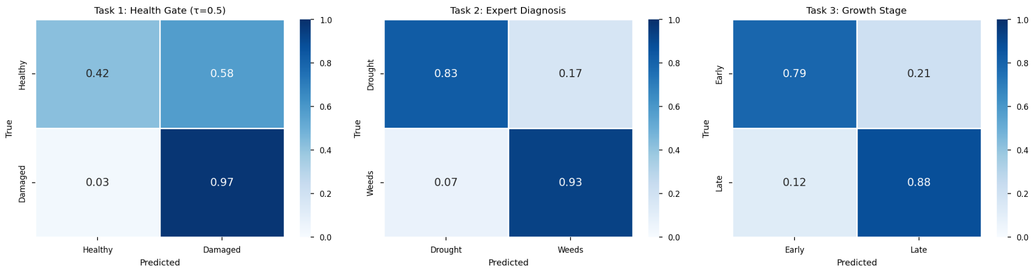

Figure 6: Normalized confusion matrices for all three tasks at $\tau = 0.5$ on the validation set. **Left (Task 1 — Health Gate):** Sentinel binary triage. Rec-Damaged = 0.97; Rec-Healthy = 0.42, reflecting the asymmetric training objective [3, 4]. **Center (Task 2 — Expert Diagnosis):** ExpertNet Drought/Weeds discrimination on truly damaged samples. F1-WED = 0.944; F1-DGT = 0.791, consistent with the 4:1 class imbalance and feature-space overlap between damage types [53, 51]. **Right (Task 3 — Growth Stage):** Sentinel auxiliary growth prediction. F1-Growth = 0.828, with Late growth (70.3% of test set) correctly classified at 0.88 [59].

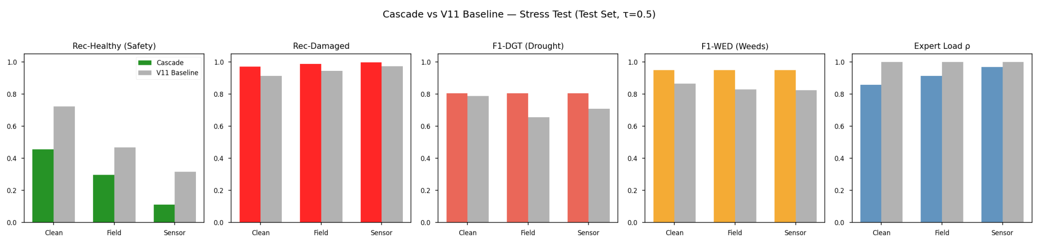

Figure 7: Cascade vs. V11 baseline robustness stress test on the test set ($n = 3{,}571$, $\tau = 0.5$). Five metrics shown across Clean, Field, and Sensor corruption conditions applied exclusively to Sentinel inputs (cascade) or uniformly to the single model (V11). **Rec-Healthy:** Cascade declines as ρ increases; V11 declines due to shared representation degradation [7]. **Rec-Damaged:** Both systems maintain high recall; cascade increases to 0.997 under Sensor noise (safe failure mode). **F1-DGT:** Cascade stable at 0.805 across all conditions; V11 drops to 0.654 under Field (−17%) and 0.708 under Sensor (−14%) relative to Clean. **F1-WED:** Cascade stable at 0.949; V11 degrades moderately [48]. **Expert Load ρ:** Cascade ρ increases from 0.856 (Clean) to 0.968 (Sensor), reflecting the safe failure mode [6]. V11 $\rho = 1.0$ throughout (no routing mechanism).

numerical comparison between them is not methodologically valid [54].

The cascade achieves higher expert-class accuracy partly because it exploits conditional label structure to reduce hypothesis complexity, trading a single harder joint problem for two simpler conditional ones [28, 24]. This reduction is not a methodological convenience — it is the architectural expression of the conditional label structure documented in Section 3: damage type is only meaningful given damage presence, and the cascade enforces this dependency at the system level. Both sets of numbers are reported transparently in Tables 4 and 5.

The comparison between ExpertNet (F1 = 0.868) and V11 (F1 = 0.777) is not directly commensurable, as ExpertNet operates on a conditional binary task while V11 performs joint three-class classification. Under evaluation alignment (Experiment 2, Section 6.7), this gap reduces to +0.008 in F1-macro — a difference too small to constitute a meaningful representational advantage on a single test set. The primary contribution of the cascade is therefore not improved representation learning, but the enforcement of safety constraints through structured inference at comparable diagnostic quality. Task decomposition is the contribution; the performance difference is its consequence.

6.7. Hypothesis Testing Experiments

Four targeted experiments evaluate the competing explanations for cascade behavior. Results are reported below; Section 7 synthesizes the verdicts.

Experiment 1 — Decision Boundary Hypothesis (H1). H1 asks whether threshold calibration of V11 alone replicates cascade performance. On the hierarchical validation set, the lowest τ satisfying Rec-D ≥ 0.95 is $\tau = 0.30$. Table 6 reports test-set performance at this operating point. The safety constraint is met under all three conditions (Rec-D ≥ 0.965), but at the cost of routing 80.8–95.6% of samples to Expert, eliminating most of the efficiency benefit. Under Field corruption, F1-DGT falls from 0.877 to 0.734 (-16%) because the flat model's shared representation propagates corruption to all outputs without isolation. **H1 is rejected.**

Table 6: V11 with threshold-calibrated routing ($\tau = 0.30$, selected on validation) evaluated on the test set (n = 3,571) under three corruption conditions (Experiment 1). The safety constraint is met but efficiency is eliminated; F1-DGT degrades under corruption due to shared-representation propagation.

Condition	Rec-D	Rec-H	F1-DGT	F1-WED	F1-Gro	ρ
Clean	0.965	0.622	0.877	0.971	0.898	0.808
Field	0.987	0.289	0.734	0.948	0.853	0.913
Sensor	0.995	0.150	0.802	0.952	0.852	0.956

Experiment 2 — Data Distribution Hypothesis (H2). H2 asks whether V11's apparent diagnostic disadvantage survives evaluation alignment. When V11 is restricted to the truly damaged test subset (n = 2,616) and its 3-class output is renormalized to a 2-class (DGT/WED) probability, F1-macro = 0.924 against ExpertNet's 0.932 — a gap of only +0.008 (Table 7). Evaluation mismatch between a 3-class model and a 2-class intrinsic evaluation accounts for most of the apparent advantage. **H2 is supported:** the cascade's diagnostic advantage is modest in representational terms; architectural isolation is a key mechanism enabling the cascade's structural advantage, not representational superiority.

Experiment 3 — Input Isolation Hypothesis (H3). H3 asks whether ExpertNet's robustness arises from structural isolation or from intrinsic learned invariance. Corruption is applied directly to ExpertNet's 256×256 inputs on all 2,616 truly damaged test images, bypassing SentinelNet entirely.

Table 7: V11 (renormalized outputs, damaged subset) vs. ExpertNet on identical samples (n = 2,616; Experiment 2). The representational gap is 0.008 F1-macro; evaluation mismatch accounts for the remainder.

Model	**F1-DGT**	**F1-WED**	**F1-macro**
V11 (renorm., damaged subset)	0.877	0.971	0.924
ExpertNet (cascade, $\tau = 0.5$)	0.893	0.971	0.932
Gap	*+0.016*	*0.000*	*+0.008*

Table 8 shows that under Sensor noise, F1-Expert falls from 0.932 to 0.745 and F1-DGT collapses from 0.893 to 0.570 (−36% relative). The stability in Table 5 is predominantly attributable to ExpertNet not receiving corrupted inputs under cascade operation; the mild Field degradation (−1.7% F1-Expert) suggests limited intrinsic resilience under weak perturbations. **H3 is strongly supported:** robustness is predominantly architectural rather than a property of the learned representation, though minor intrinsic robustness cannot be ruled out under mild corruption.

Table 8: ExpertNet evaluated with corruption applied directly to its inputs (SentinelNet bypassed; Experiment 3). Contrast with Table 5 where corruption applies to Sentinel inputs only. F1-DGT collapses −36% under Sensor noise, confirming that isolation drives the robustness.

Condition	**F1-Expert**	**F1-DGT**	**F1-WED**
Clean	0.932	0.893	0.971
Field	0.916	0.868	0.963
Sensor	0.745	0.570	0.920

Experiment 4 — Control Parameter Hypothesis (H4). H4 asks whether τ transfers across environments without local recalibration. Table 9 reports Rec-D and ρ on validation and test splits for six operating points. The maximum Rec-D gap is $\Delta = 0.013$ at $\tau = 0.7$; the maximum ρ gap is $\Delta = 0.011$ at the same point. Both remain below 0.02 throughout the sweep. **H4 is supported:** τ transfers reliably across held-out splits of the same distribution; practitioners may calibrate on a validation set and deploy without re-tuning under matched conditions. Cross-distribution transfer — different geography, crop type, or imaging device — would require re-calibration and is not tested here.

Table 9: τ transfer: validation vs. test (n = 3,571 each; Experiment 4). Δ denotes the absolute split difference. Maximum Δ Rec-D = 0.013 at $\tau = 0.7$; τ calibrates stably within same-distribution splits.

τ	**Rec-D val**	**Rec-D test**	Δ	ρ **val**	ρ **test**	Δ
0.3	1.000	0.999	0.001	0.969	0.970	0.000
0.4	0.995	0.996	0.001	0.928	0.931	0.004
0.5	0.977	0.978	0.000	0.864	0.864	0.000
0.6	0.938	0.932	0.006	0.771	0.765	0.006
0.7	0.851	0.837	**0.013**	0.659	0.648	**0.011**
0.8	0.680	0.673	0.006	0.508	0.504	0.004

Experiment 5 — Oracle Routing Upper Bound. The oracle routes all truly damaged samples directly to ExpertNet and predicts Healthy for all healthy samples, eliminating all routing error. Table 10 shows that cascade ($\tau = 0.5$) and oracle produce identical F1-Expert, F1-DGT, and F1-WED (gap = 0.000). This arises from the intrinsic evaluation protocol: ExpertNet is evaluated on the full damaged subset regardless of routing, so routing quality is not captured by Expert metrics alone. The oracle analysis confirms that, under the intrinsic evaluation protocol, *classification* performance defines the measurable system ceiling: improving ExpertNet's diagnostic accuracy yields greater gain than improving SentinelNet's routing precision *as measured here*. In deployment, where routing errors directly determine which samples reach ExpertNet, routing quality would also surface in end-to-end metrics.

Table 10: Oracle routing upper bound vs. cascade at $\tau = 0.5$ on the test set (Experiment 5). Under the intrinsic evaluation protocol, oracle and cascade produce identical Expert metrics (gap = 0.000). The oracle is an analytical bound, not a deployable configuration.

Routing	**Rec-D**	**Rec-H**	ρ	**F1-Exp**	**F1-DGT**	**F1-WED**
Oracle (perfect)	1.000	1.000	0.733	0.932	0.893	0.971
Cascade ($\tau = 0.5$)	0.978	0.449	0.864	0.932	0.893	0.971

Experiment 6 — End-to-End Routing Bias Analysis. The intrinsic evaluation protocol (Experiments 1–5) evaluates ExpertNet on the full truly-damaged subset regardless of routing. Experiment 6 closes this gap by running the cascade end-to-end: SentinelNet routes each sample, and ExpertNet processes only what it actually receives. Table 11 reports routing composition and performance at five operating points.

At $\tau = 0.5$, 526 truly healthy samples (55.1% of all healthy crops) are routed to ExpertNet — all of them receiving a false damage diagnosis, since ExpertNet has no Healthy class. Of these, 90% are misclassified as WED (weed presence) and 10% as DGT (drought), a systematic bias consistent with healthy crops sharing more visual features with the weed-affected class. The protocol gap between intrinsic and TP-routed Expert F1 is -0.0004 (below measurement precision), confirming that the intrinsic evaluation protocol is not artificially inflated relative to correctly-routed performance. The end-to-end 3-class F1-macro at $\tau = 0.5$ is 0.767, against the intrinsic F1-Expert of 0.932 — the gap reflects false positive routing, not ExpertNet's classification quality.

Table 11: End-to-end routing evaluation across five operating points (Experiment 6). TP, FP, FN, TN denote routing outcomes. FP-rate-H: fraction of healthy crops misrouted to Expert; all receive a false damage label. Protocol gap: intrinsic F1-Expert minus TP-routed F1-Expert; near-zero confirms the intrinsic protocol is accurate. F1-E2E: true 3-class macro F1 (Healthy/DGT/WED).

τ	TP	FP	FN	TN	**FP-rate-H** (% healthy)	**Gap** (intr−TP)	**F1-E2E** (3-class)
0.3	2613	849	3	106	89%	+0.000	0.618
0.4	2606	720	10	235	75%	−0.000	0.690
0.5	2558	526	58	429	55%	−0.000	0.767
0.6	2438	293	178	662	31%	+0.000	0.821
0.7	2190	124	426	831	13%	−0.001	0.811

This result has a design implication: to reduce false damage diagnoses for healthy crops, τ should be raised above 0.5. At $\tau = 0.6$, the FP route rate drops from 55% to 31% and end-to-end F1-macro improves from 0.767 to 0.821 — but at the cost of Rec-D falling from 0.978 to 0.932, below the 0.95 safety floor. The operating point is a genuine trade-off between missed damage recall and healthy crop misdiagnosis.

Experiment 7 — Both-Corrupted Stress Test. To close the comparison asymmetry in Table 5 (cascade partially corrupted vs V11 fully corrupted), Experiment 7 applies the same corruption to both Sentinel (224×224) and Expert (256×256) inputs simultaneously. Table 12 reports all seven scenarios. Three findings emerge.

First, the isolation boundary remains effective under combined corruption. Both-Field and Expert-only-Field produce nearly identical Expert metrics (F1-Exp 0.914 vs 0.913, F1-DGT 0.865 vs 0.864). Both-Sensor and Expert-only-Sensor are similarly indistinguishable (F1-Exp 0.745 vs 0.744, F1-DGT 0.569 vs 0.568). Sentinel corruption adds no marginal Expert degradation because Sentinel and Expert inputs are independently preprocessed: even when both stages are corrupted, the Expert's performance is determined solely by what reaches its input, not by what happened upstream.

Second, pipeline failures are additive, not multiplicative. The interaction term is +0.001 for both Field and Sensor conditions — below measurement precision. There is no compound failure mode: corrupting Sentinel and Expert simultaneously produces exactly the sum of corrupting each independently.

Third, when the pipeline assumption fully fails (Both Sensor), the cascade provides no robustness advantage: F1-DGT collapses to 0.569 (−36%) and efficiency is eliminated ($\rho = 0.923$). The system's safety properties require maintaining ExpertNet input quality. If that condition holds, the isolation delivers the robustness shown in Table 5; if it fails, performance matches the Expert-only corrupted scenario.

Table 12: Both-corrupted stress test at $\tau = 0.5$ (Experiment 7). Seven scenarios compare Sentinel-only, Expert-only, and simultaneous corruption. Interaction terms (Both minus additive sum) are +0.001 for Field and Sensor, confirming that failures are additive and the isolation boundary contains Sentinel-stage corruption even under simultaneous Expert corruption. All metrics on test set ($n = 3{,}571$) using the intrinsic evaluation protocol.

Sentinel	**Expert**	**Rec-D**	**Rec-H**	ρ	**F1-Exp**	**F1-DGT**	**F1-WED**
Clean	Clean	0.978	0.449	0.864	0.932	0.893	0.971
Field	Clean	0.989	0.269	0.920	0.932	0.893	0.971
Clean	Field	0.978	0.449	0.864	0.913	0.864	0.962
Field	**Field**	0.991	0.276	0.919	**0.914**	**0.865**	**0.962**
Sensor	Clean	0.976	0.212	0.926	0.932	0.893	0.971
Clean	Sensor	0.978	0.449	0.864	0.744	0.568	0.920
Sensor	**Sensor**	0.975	0.220	0.923	**0.745**	**0.569**	**0.920**

7. Discussion

The results reported in Section 6 are consistent with the paper's organizing hypothesis, but consistency is not the same as confirmation, and the discussion that follows tries to hold that distinction seriously [13, 8].

Four competing explanations for the observed cascade behavior deserve explicit evaluation. The *decision boundary hypothesis* asks whether threshold calibration of a flat model alone replicates cascade behavior. The *data distribution hypothesis* asks whether the diagnostic advantage survives evaluation alignment — that is, whether V11's representational quality on the damaged subset is genuinely inferior or whether the gap reflects evaluation mismatch between a 3-class model and a 2-class intrinsic evaluation. The *input isolation hypothesis* asks whether robustness arises from structurally decoupling the diagnostic model from corrupted upstream inputs, rather than from learned representational invariance. The *control parameter hypothesis* asks whether τ transfers across environments or requires local recalibration. Section 6.7 reports five targeted experiments that evaluate each hypothesis directly. H1 is rejected: threshold calibration of a flat model satisfies the safety constraint only at the cost of eliminating efficiency, and the flat representation still propagates corruption to diagnostic outputs. H2 is supported: under fair evaluation on identical samples, the representational gap shrinks to 0.008 F1-macro, confirming that evaluation mismatch drove most of the apparent advantage. H3 is strongly supported: ExpertNet degrades severely when directly corrupted (F1-DGT -36% under Sensor noise), confirming that isolation is the predominant source of robustness; limited intrinsic resilience may persist under mild Field corruption (-1.7% F1-Expert). H4 is supported: the maximum Rec-D transfer gap is 0.013 across same-distribution splits, confirming stable calibration within distribution; cross-distribution transfer is not tested. The oracle analysis confirms that, under the intrinsic evaluation protocol, classification performance defines the measurable system ceiling.

Taken together, the five experiments support a consistent framing: system performance is governed by a joint objective over routing accuracy (Rec-D, ρ) and conditional classification performance (F1-DGT, F1-WED) that cannot be reduced to a single classifier metric. The two objectives are coupled in deployment but partially decoupled by the intrinsic evaluation protocol, which is why the oracle gap is zero while routing errors remain present.

7.1. τ as an Operational Decision

The threshold τ is not a tuning parameter in the conventional sense. It does not affect model weights, training dynamics, or ExpertNet's learned parameters or its intrinsic performance under a fixed evaluation distribution. As shown in Table 4 and Figure 5, F1-Expert remains fixed at 0.868 and F1-Growth at 0.828 across all six operating points $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$. What τ controls is the composition of the routing decision: how much of the input stream is sent to the Expert, and at what cost to Rec-Damaged [3, 4]. This is an operational decision that belongs to the deploying organization, not a modeling decision that belongs to the training process [10, 2].

Two operating points emerge from the data as natural anchors. At $\tau = 0.5$, Rec-Damaged = 0.974 and $\rho = 0.868$, meaning 13.2% of samples bypass the Expert entirely (Table 4). This is the safety-first configuration: the recall constraint is satisfied with margin, at the cost of routing the majority of the input stream to ExpertNet. At $\tau = 0.6$, Rec-Damaged = 0.929 and $\rho = 0.774$, achieving 22.6% expert load reduction with F1-bin = 0.804 — the best binary classification performance across the sweep [41, 45]. Whether $\tau = 0.5$ or $\tau = 0.6$ is appropriate depends on the operator's tolerance for missed damage cases relative to Expert model capacity — a decision the system supports but cannot make [1, 60].

To make the trade-off concrete: at $\tau = 0.5$, the cascade routes 86.8% of submitted images to ExpertNet, corresponding to an expert load reduction of 13.2% compared to routing all images. For a deployment processing 10,000 images per season, this corresponds to approximately 1,320 fewer expert reviews — a direct reduction in operational cost without compromising the Rec-Damaged ≥ 0.95 safety floor. At $\tau = 0.6$, the reduction increases to 22.6%, or approximately 2,260 fewer reviews per 10,000 images, at the cost of Rec-Damaged $= 0.929$. These figures are deployment-scale quantities, not percentage points: they translate directly into agronomist workload and insurance processing capacity for the operator.

At $\tau = 0.5$, 57.6% of truly healthy plots are routed to ExpertNet as potentially damaged, implying that an insurance organization processing 10,000 plots per season would generate approximately 2,723 unnecessary expert reviews alongside the 1,320 avoided through expert load reduction. Increasing the threshold to $\tau = 0.6$ improves specificity (Rec-Healthy $= 0.650$), reducing unnecessary reviews to 35% of healthy plots. More generally, given a relative cost ratio between false negatives and false positives, the optimal operating threshold τ can be selected by minimizing expected decision cost over a validation set, enabling deployment-specific calibration according to institutional risk preferences without retraining.

This control is meaningful under stable deployment conditions; it requires revalidation on locally representative data as Sentinel probability calibration drifts, and the operating characteristics associated with a fixed τ should not be assumed to transfer across geographies or sensor environments without empirical verification (Section 8).

At the extremes, the Pareto frontier becomes uninformative. At $\tau = 0.3$,

$\rho = 0.973$ and Rec-Damaged $= 0.997$ — virtually no damage is missed, but the Expert processes nearly everything, eliminating the cascade's efficiency rationale [46]. At $\tau = 0.8$, $\rho = 0.500$ and Rec-Damaged $= 0.660$ — half the input stream bypasses the Expert, but the probability of missing real damage is unacceptably high for an insurance-linked workflow [2, 10]. The usable operating range is roughly $\tau \in [0.5, 0.7]$, where the Pareto frontier offers meaningful trade-offs between safety and efficiency, as visualized in Figure 5. While thresholding can be applied to flat classifiers [35, 36], the resulting trade-off is not structurally decoupled from representation learning and, as observed in Section 6.5, degrades more sharply under distribution shift in this setting [7, 8].

7.2. Robustness: What the Architecture Provides and What It Requires

ExpertNet does not exhibit robustness to corrupted inputs when evaluated in isolation (Experiment 3): under direct Sensor noise, F1-DGT collapses from 0.893 to 0.570, a 36% relative degradation. The stability reported in Table 5 arises entirely from the preprocessing pipeline delivering clean images to ExpertNet regardless of Sentinel-input corruption. Robustness in this system therefore arises from architectural isolation and error containment, not from learned invariance within the model itself.

Experiment 7 (Section 6.7) resolves the asymmetric-comparison concern in Table 5: when corruption is applied to both stages simultaneously, failures are additive (interaction = +0.001, below measurement precision) and the isolation boundary remains effective. Both-corrupted Field and Sensor scenarios are empirically indistinguishable from their Expert-only counterparts, because Sentinel and Expert inputs are independently preprocessed. The cascade does not amplify simultaneous failures.

This result sharpens the precision of the robustness claim. The architecture provides one guarantee: if ExpertNet's input quality is maintained, ExpertNet's outputs are unaffected by what happens at the Sentinel stage. Experiment 7 confirms this guarantee holds even when Sentinel is simultaneously corrupted. What the architecture cannot provide is robustness against ExpertNet input corruption: under Sensor noise applied to ExpertNet's inputs, F1-DGT collapses from 0.893 to 0.569 (-36%) whether or not Sentinel is also corrupted. The system's robustness depends entirely on maintaining ExpertNet input quality at deployment — a pipeline requirement, not a model property.

Practically, this means the robustness guarantee in Table 5 should be read as conditional: it holds under the deployment assumption that ExpertNet receives server-side preprocessed images while Sentinel inputs arrive from heterogeneous field devices. If transmission artifacts, lossy compression, or device-side preprocessing failures reach ExpertNet inputs, the system degrades to the Expert-only corrupted scenario shown in Table 12 — a condition operators should monitor and guard against by design.

7.3. The Safe Failure Mode and Its Cost

The most practically significant finding in the robustness stress test (Table 5; Figure 7) is not that the cascade outperforms V11 under corruption — though it does, substantially — but that its failure mode is structurally safe [6, 4]. In effect, the cascade converts input uncertainty into routing escalation rather than prediction error [13]. Under sensor noise ($\sigma = 0.08$ additive Gaussian, applied post-normalization), Rec-Healthy drops to 0.110 and ρ increases to 0.968 at $\tau = 0.5$. The system is routing nearly everything to ExpertNet. This is not good performance. But it is a safe failure: the Sentinel, uncertain under degraded inputs [50], escalates rather than commits. Rec-Damaged increases to 0.997 under sensor noise because the Sentinel's bias toward routing removes the possibility of a false Healthy prediction on damaged crops. This safety guarantee, however, is contingent on ExpertNet maintaining its diagnostic reliability under the same shift conditions — a property that holds here because the Expert always receives clean images, but which should not be assumed to generalize to pipeline configurations where Expert inputs are also corrupted [13, 8].

The cost of this safe failure mode is total elimination of the cascade's efficiency benefit. At $\rho = 0.968$, the Expert processes 96.8% of the input stream — functionally equivalent in routing volume to running ExpertNet on everything, though not necessarily identical in computational overhead due to pipeline structure [41, 46]. The cascade's 13–22% expert load reduction, documented under clean conditions, disappears entirely under severe sensor degradation (Table 5). Operators should plan for this: a system that claims computational efficiency under field conditions must account for the full distribution of sensor quality it will encounter [10, 9], not just the clean-condition baseline.

The efficiency benefit observed under clean conditions — a 13.2% reduction in expert reviews at $\tau = 0.5$ — collapses under severe input degradation ($\rho = 0.968$ under Sensor noise), where most samples are escalated to ExpertNet. This behavior is intentional: the cascade trades efficiency for safety under uncertainty, reverting toward full expert coverage rather than producing confident misclassifications. Importantly, this conditional computation differs from deploying ExpertNet alone, which would incur full computational and operational cost across all inputs regardless of input quality. The cascade therefore acts as an adaptive routing mechanism, preserving baseline safety guarantees while enabling efficiency gains when input conditions permit.

V11 fails differently under the same corruption. Rec-Healthy drops from

0.722 to 0.316 under sensor noise — a 56% reduction — and F1-DGT drops from 0.787 to 0.654 under field corruption (Table 5 and Figure 7). These degradations occur simultaneously and without any compensating mechanism. V11 has no architectural isolation; corruption that enters the shared backbone [55] propagates to all task heads equally [7, 48]. The cascade's advantage is not immunity to corruption — the Sentinel's routing is clearly disrupted — but that the disruption is contained to the routing decision rather than to the diagnostic output. Viewed more broadly, the cascade reframes classification as a form of risk allocation [3, 5], where uncertain cases are escalated rather than forced into potentially incorrect predictions.

7.4. Drought Recall as an Unresolved Problem

F1-DGT = 0.791 is the result that should give practitioners the most pause. The confusion matrix for Task 2 (Figure 6, center panel) shows the Expert correctly classifying 83% of Drought cases and misclassifying 17% as Weeds. In an insurance context where the agronomic response to drought stress — irrigation, replanting, input substitution — differs materially from the response to weed competition [10, 2], a 17% misclassification rate on the minority damage class is not a minor modeling imperfection. It is a diagnostic reliability problem [51, 21].

FocalLoss with $\gamma = 2.0$ [53] partially addresses the 4:1 Weeds/Drought imbalance, and the Expert's convergence dynamics (Figure 4) show F1-WED rising steeply and stabilizing near 0.95 by epoch 5 while F1-DGT plateaus near 0.79 after epoch 5 and does not improve thereafter. This is not a training convergence failure [57]. This suggests a degree of non-separability between drought and weed stress in RGB feature space at certain growth stages — leaf yellowing, stunted growth, and canopy thinning occur in both conditions, particularly when canopy coverage is low and distinguishing features are less visually pronounced [18, 22]. While feature-space overlap is a plausible explanation, alternative factors such as label noise or stage-dependent annotation ambiguity in the mixed manual-automatic labeling pipeline [9, 30] cannot be excluded without further analysis. Resolving this gap may require substantially more Drought-class examples [34], or auxiliary inputs — multispectral bands, temporal sequences capturing damage progression [21] — that a single RGB smartphone image cannot provide [9, 10].

7.5. Rec-Healthy at $\tau = 0.5$: A Feature, Not a Bug

Rec-Healthy = 0.424 at $\tau = 0.5$ (Figure 6, left panel) requires direct explanation rather than defensive qualification. The Sentinel was trained under a constraint that prioritizes Rec-Damaged ≥ 0.95, with Rec-Healthy ≥ 0.40 as a secondary floor [3, 4]. The checkpoint selected at epoch 4 — Safety Score = 0.8532 (Figure 3) — satisfies both constraints. The consequence is a classifier that aggressively routes toward damage: 57.6% of Healthy crops are misrouted to ExpertNet as potentially damaged [33, 36]. This cost is not incidental — it is the direct consequence of enforcing the Rec-Damaged constraint, and it was accepted by design [3, 5].

The practical implication is that operators deploying at $\tau = 0.5$ should expect ExpertNet to process a substantial proportion of genuinely healthy crops [10]. At $\tau = 0.6$, Rec-Healthy improves to 0.650 — still not high in absolute terms, but meaningfully better — at the cost of Rec-Damaged falling to 0.929. Whether 0.929 is an acceptable safety floor is context-dependent [2, 60]. In a high-stakes insurance payout setting where every missed damage case has direct financial consequences for a vulnerable farmer [1], it may not be. In a preliminary screening context where a human agronomist reviews ExpertNet outputs before any payout decision [10], it may be entirely reasonable. For practitioners deploying in field conditions where expert review bandwidth is limited [10], $\tau = 0.6$ or $\tau = 0.7$ is likely the more realistic operating point: both deliver substantially higher Rec-Healthy (0.650 and 0.816 respectively) while maintaining Rec-Damaged above operationally meaningful thresholds [2, 1]. $\tau = 0.5$ should be reserved for contexts where the cost of any missed damage case is the dominant operational constraint.

Taken together, the system's behavior is governed by the interaction between training constraints, threshold calibration, and input uncertainty [50, 13], rather than by model accuracy alone. This is the distinguishing property of a safety-constrained system: its deployment behavior is as much a function of its design objectives [3, 4] as of its learned representations [55, 24].

7.6. Toward Adaptive Routing

The fixed-threshold design of τ is a deliberate simplicity choice — it requires no additional modeling, introduces no trainable parameters at deployment time, and is fully interpretable [6]. But it is not the only possible design. Two extensions are natural.

The first is uncertainty-aware routing, where the Sentinel's routing decision is based on predictive entropy rather than a fixed probability threshold. A sample with $P(\text{Damaged}|\mathbf{x}) = 0.55$ and high entropy — indicating genuine model uncertainty [13] — warrants Expert consultation more strongly than a sample with the same probability but low entropy. Fixed-threshold routing treats these identically; entropy-based routing would not [6, 43]. This could reduce ρ under clean conditions without sacrificing Rec-Damaged, by routing only genuinely ambiguous cases rather than all cases above a probability cutoff [41, 44]. However, the reliability of uncertainty estimates under distribution shift remains an open question [13, 50] — calibration may degrade precisely under the conditions where routing decisions become most critical [8], inverting the intended behavior at the worst possible moment.

The second is adaptive τ calibration, where τ is adjusted dynamically based on observed batch statistics or domain shift indicators [7, 47]. The threshold drift from $\tau = 0.56$ at epoch 4 to $\tau = 0.19$ at epoch 30 (Figure 3, right panel) is evidence that optimal τ is sensitive to the model's confidence calibration [50]. A deployment system that monitors routing rate ρ and adjusts τ to maintain a target ρ could preserve efficiency under moderate corruption without retraining [43, 45]. Both extensions remain directions for future work; the current system provides the baseline against which they should be evaluated.

8. Limitations

The limitations of CascadeCropNet are not peripheral edge cases — they are structural properties of the design that follow directly from the architectural choices and training objectives described in Sections 4 and 5. Stating them as such is more useful than treating them as caveats.

The cascade introduces a hard error boundary at the triage stage. False negatives at the Sentinel — truly damaged crops assigned $P(\text{Damaged}|\mathbf{x}) < \tau$ and therefore predicted Healthy — never reach ExpertNet. In a flat classifier, errors are not gated by an irreversible intermediate decision, and all samples remain available to downstream representations [24, 28]. The cascade localizes failure at the routing stage, making Sentinel false negatives both identifiable in structure and lost to the system within the current routing configuration [6]. No downstream correction mechanism exists within the pipeline. At $\tau = 0.5$, this means 2.6% of truly damaged crops are missed by the system entirely (Table 4) — a figure that should be understood not as an acceptable error rate but as a hard floor imposed by the Sentinel's recall at this operating point [3, 4]. In an insurance-linked context [2, 1], each such case represents a farmer who sustained real losses and received no payout.

Failure in the cascade is asymmetric in a deeper sense than the false negative rate alone captures. While not evaluated empirically in this work, under more severe distribution shift affecting Sentinel recall — geographic transfer to regions with substantially different damage presentations [11, 12], or sensor degradation beyond the corruption levels tested here [7] — the cascade may underperform flat models entirely, as Sentinel false negatives accumulate faster than the architectural isolation property can compensate [8, 47]. More precisely: because routing decisions are irreversible, degradation in Sentinel recall disproportionately impacts system performance, and improvements in downstream Expert accuracy cannot compensate for degraded Sentinel recall [6, 5].

Drought classification remains an open problem within the current system. F1-DGT = 0.791 is insufficient for treatment-level diagnostic reliability in contexts where the response to drought stress differs materially from the response to weed competition [10, 2]. The 4:1 Weeds/Drought class imbalance [33, 34] is a primary driver, though feature-space overlap between drought and weed stress in RGB feature space at certain growth stages [18, 22], and potential label noise or stage-dependent annotation ambiguity in the mixed manual-automatic labeling pipeline [9, 30], likely contribute as well. FocalLoss with $\gamma = 2.0$ [53] partially addresses the imbalance but does not resolve the underlying non-separability [52]. Per-class oversampling [34], synthetic augmentation targeting Drought presentations [49], or auxiliary inputs beyond RGB [21] are candidate directions; none has been evaluated in this work.

Expert performance is reported on truly damaged samples under the intrinsic evaluation protocol described in Section 5. Experiment 6 (Section 6.7) quantifies the consequence of this choice. The intrinsic–TP-routed protocol gap is -0.0004 at $\tau = 0.5$, confirming that the intrinsic protocol accurately reflects ExpertNet's performance on correctly-routed samples. However, the end-to-end 3-class F1-macro is 0.767 against the intrinsic F1-Expert of 0.932 — a gap of 0.165 that arises entirely from false positive routing. At $\tau = 0.5$, 55% of healthy crops are routed to ExpertNet and receive a false damage diagnosis; 90% of these false diagnoses are WED (weed presence), reflecting a systematic visual similarity between healthy crops and the weed-affected class. Practitioners should monitor end-to-end 3-class accuracy, not intrinsic Expert F1, when evaluating deployed system performance.

A structural training asymmetry also warrants acknowledgement. V11 is trained on all three classes (Healthy, DGT, WED) using the full training set; ExpertNet is trained only on the damaged subset (Section 4.5). This is not purely an architectural advantage — it is a form of curriculum learning and conditional filtering that may improve performance independently of the cascade structure. Experiment 2 isolates the representational gap to +0.008 F1-macro under fair evaluation, but does not fully disentangle training distribution from architecture. A fully controlled comparison would require training a flat model on the damaged subset only, which has not been conducted.

The evaluation is conducted on a single dataset from a specific geographic and agronomic context — smallholder maize farms in eight Kenyan counties during 2020–2021. The cascade architecture and the τ-based control mechanism are general, but the specific performance values reported (Rec-Damaged = 0.974, F1-Expert = 0.868, Safety Score = 0.8532) reflect this context. Transfer to other crops, geographies, or smartphone camera distributions would require local revalidation of τ and potentially retraining of both models. F1-DGT = 0.791 specifically limits treatment-level deployment for drought diagnosis — this is a structural limitation of the diagnostic stage under the current class imbalance and RGB feature space, not of the cascade architecture itself.

The system's operational behavior depends critically on the calibration of Sentinel output probabilities and the selection of τ on validation data [50, 13]. Under deployment conditions where probability calibration shifts — due to geographic transfer [11], sensor change [7], or distributional drift over time [47] — the operating characteristics associated with a fixed τ may no longer hold. A τ selected to achieve Rec-Damaged = 0.974 on the Kenyan validation set [9] may produce substantially different recall under a different deployment distribution [8]. Periodic revalidation of τ on locally representative data is advisable [10, 60], and the τ-as-Pareto-parameter framing in Section 4.1 is designed precisely to make this recalibration tractable without retraining. This does not eliminate the dependency on local calibration data [50].

While the optimal threshold τ may vary under distribution shift and requires local revalidation, the trade-off structure between Rec-Healthy and Rec-Damaged remains monotonic across all evaluated settings (Table 4). This ensures that threshold adjustment is predictable and preserves the intended safety–efficiency trade-off, even when deployment conditions differ from the training distribution.

A methodological asymmetry in the evaluation design warrants explicit acknowledgement as a structural limitation rather than a footnote. The threshold sweep — from which the two primary operating points ($\tau = 0.5$, $\tau = 0.6$) are derived — is conducted on the validation set, while the robustness stress test is conducted on the held-out test set. This means the paper's two central results are not directly comparable as absolute numbers: the safety performance figures (Section 6.2) and the robustness figures (Section 6.5) come from different data partitions. Harmonizing both evaluations to a single held-out split — by running either the threshold sweep on the test set or the stress test on the validation set — would resolve this inconsistency and should be prioritized in follow-on work [35].

More broadly, the architecture imposes a structural coupling between safety and efficiency: the recall constraint that enforces the safety floor simultaneously drives aggressive routing [3, 4], and any improvement in Rec-Damaged at a given τ comes at the cost of increased expert load ρ [6, 41]. This is not a property that can be engineered away within the current design — it is a consequence of the constrained optimization formulation in Section 4.1 [5]. Relaxing the coupling would require either a more expressive Sentinel backbone [61, 62] that achieves better class separation, or a routing mechanism that conditions on richer signals than scalar damage probability alone [43, 44].

The cascade architecture represents one concrete instantiation of enforcing safety constraints through structured inference. Alternative approaches, such as multi-head or hierarchical loss formulations, may achieve similar predictive performance but do not provide the same structural separation of triage from diagnosis at inference time. In contrast, the cascade enforces safety through conditional execution, ensuring that uncertain or high-risk cases are systematically escalated. Empirical comparison with such alternatives is left for future work.

The cascade provides structural interpretability: the decision pathway is explicit and traceable, with triage and diagnosis separated into distinct, inspectable stages. It does not provide epistemic interpretability. The system does not quantify predictive uncertainty or indicate when its outputs should be distrusted, which limits its use in risk-sensitive decision workflows such as agricultural insurance where operators need to know not only *where* a decision occurred but *how confident* the system was when it made it. Deterministic routing policies of this form make failure analyzable but not detectable at inference time, as the system lacks mechanisms to signal when its predictions may be unreliable. Uncertainty-aware routing, discussed in Section 7.6, is one candidate direction toward closing this gap.

9. Conclusion

This work addresses a fundamental mismatch between the asymmetric cost structure of insurance-linked crop damage assessment and the objectives typically used to train automated classification systems [3, 4, 10]. Prior work has formalized asymmetric risk through cost-sensitive learning [3, 32] and constrained classification [4, 5] — the theoretical apparatus exists. What has received less attention is how these ideas translate into systems that maintain safety-aligned behavior when input distributions shift at deployment [7, 8, 13]. This paper proposes an architectural response: a hierarchical cascade that explicitly separates binary triage from fine-grained diagnosis, enforces a recall constraint at the level of model selection during training, and structurally isolates diagnostic inference from corruption affecting triage inputs.

Empirically, the cascade exhibits strong robustness under controlled corruption applied to Sentinel inputs. Across all tested conditions, ExpertNet maintains stable diagnostic performance — F1-Expert = 0.877, F1-DGT = 0.805, F1-WED = 0.949 — while the flat baseline degrades by 17 to 35 percent across the same metrics under the same conditions (Table 5). These results suggest that the observed robustness is closely linked to the architectural isolation between stages, which constrains how perturbations propagate through the system [7, 48]. This interpretation is based on the corruption regimes evaluated in this study and does not establish invariance under broader forms of distribution shift [47, 8], including geographic transfer beyond the Kenyan training distribution [9] or sensor configurations not represented in the test set [10].

The routing threshold τ provides a practical mechanism to adapt system behavior at deployment without retraining [6, 41]. By adjusting τ, operators can navigate the trade-off between expert workload and recall of damaged crops along a Pareto frontier whose axes are directly interpretable: expert load ρ and Rec-Damaged [5, 4]. At $\tau = 0.5$, Rec-Damaged = 0.974 with a 13.2% reduction in expert inference volume; at $\tau = 0.6$, Rec-Damaged = 0.929 with a 22.6% reduction (Table 4). Under corruption, the system exhibits a conservative failure mode — routing becomes more aggressive, increasing expert load while preserving diagnostic quality [13, 50] — consistent with the safety requirements of workflows where missed damage carries greater operational cost than unnecessary review [2, 1, 10].

Several limitations remain and should be understood as structural rather than incidental. Drought classification at F1-DGT = 0.791 is insufficient for reliable treatment-level diagnosis, likely reflecting a combination of class imbalance [33, 34], visual overlap between drought and weed stress in RGB feature space [18, 22], and potential annotation noise in the mixed labeling pipeline [9, 30]. The mapping between τ and system recall depends on Sentinel probability calibration [50], which may shift under geographic transfer [11] or changing sensor conditions [7]; deployment in new regions requires local revalidation to maintain the intended safety guarantees [60, 10]. The intrinsic-deployed performance gap for ExpertNet — evaluated here on truly damaged samples rather than sentinel-routed samples (Section 5) — remains empirically unquantified and constitutes an open problem for follow-on work [13, 54].

More broadly, this work suggests that incorporating decision structure at the architectural level may offer a more reliable pathway to handling asymmetric risk under distribution shift than loss-based approaches alone [3, 4, 5]. The cascade is one concrete instantiation of this idea, grounded in the conditional label structure of the Eyes on the Ground dataset [9] and the operational requirements of picture-based agricultural insurance [10, 2, 1]. Whether it generalizes across domains and task structures where similar cost asymmetries and distribution shift conditions hold [7, 8] is a question that the results here motivate but do not answer.

CRediT authorship contribution statement

José T. M. Hagbe: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization. **Kawsar Gounou:** Writing — review & editing, Validation. **Songbian Zimé:** Writing — review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethical statement

This study uses the publicly available Eyes on the Ground dataset [9], collected under informed consent as part of a picture-based insurance protocol [10]. No new human or animal subjects were involved in this research.

Data availability

The Eyes on the Ground dataset is publicly available on Radiant MLHub at `https://doi.org/10.34911/rdnt.1bs2jw` [9] under a CC-BY-SA-4.0 license. Model checkpoints and evaluation scripts are available from the corresponding author upon reasonable request.

Acknowledgements

The Eyes on the Ground dataset was produced by ACRE Africa, the International Food Policy Research Institute, the Lacuna Fund, KALRO, BlueGreen Labs, and Dvara E-Registry [9]. The authors thank the smallholder farmers who contributed images under the picture-based insurance protocol [10].

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